IF001

IF001-CL201

Protocol Title:	A Phase 2 Single-Arm, Multicenter, Exploratory Safety and Efficacy Study of IF001 to **Re**move **L**ong-Term **I**mmunosuppr**e**ssion **f**rom Prior Living Donor Kidney Transplant Recipients (RELIEF)
Indication Studied:	Prevention of living donor kidney allograft rejection in adult recipients following discontinuation of pharmacologic immunosuppression
Phase:	2
IND Number:	16834
Sponsor Address:	ImmunoFree, Inc. 585 West Putnam Avenue Greenwich CT 06830 United States
Protocol Version/Date:	1.0, DD MMM 2025 DRAFT

This study will be conducted in compliance with the protocol, Good Clinical Practice, and all other applicable regulatory requirements, including the archiving of essential documents.

regulations. In any event, persons to whom the information is disclosed must be informed that the information is privileged or confidential and may not be further disclosed by them. These restrictions on disclosure will apply equally to all future information supplied to you which is indicated as privileged or confidential.

Synopsis

Name of Sponsor/Company: ImmunoFree, Inc.	
Name of Investigational Product: IF001	
Name of Active Ingredient: Investigational cryopreserved allogeneic somatic cell therapy, derived from donor mobilized peripheral blood mononuclear cells	
Title of Study: A Phase 2 Single-Arm, Multicenter, Exploratory Safety and Efficacy Study of IF001 to **Re**move **L**ong-Term **I**mmunosuppr**e**ssion **f**rom Prior Living Donor Kidney Transplant Recipients (RELIEF)	
Study center(s): Multiple centers in the United States	
Study period (years): Estimated date first patient enrolled: Day Month Year Estimated date last participant completed: Day Month Year	**Phase of development:** 2
Objectives: <u>Safety</u> Primary: ● Determine the safety and tolerability of IF001 in previously transplanted recipients of a kidney from a living donor Secondary: ● Characterize the incidence and severity of acute graft-versus-host disease (GVHD) by Day 120 and of chronic GVHD by one-year post-IF001 infusion ● Characterize the incidence of BK viremia, viruria, infection, and nephropathy by 15 months post-IF001 infusion ● Characterize the incidence of proteinuria, neurotoxicity, anemia, diabetes, hypertension, dyslipidemia, major adverse cardiovascular events, and malignancies ● Describe: o The time from IF001 infusion to neutrophil recovery (>500/µL) and platelet recovery (>20K/µL) o The number of blood component transfusions o The incidence of engraftment syndrome o The incidence of primary and secondary hematopoietic graft failure o The incidence and severity of bacterial, fungal, and viral infections <u>Efficacy</u>: Primary: ● Evaluate the efficacy of IF001 in preventing graft loss and biopsy-proven acute rejection (BPAR) following removal of pharmacologic immunosuppression (IS) <u>Secondary</u>:	

- Evaluate changes in renal function (estimated glomerular filtration rate [eGFR] by Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI 2021) from Day −9, prior to the first dose of transplantation conditioning, to Month 15
- Evaluate the time to the event for the composite of BPAR, graft loss, death, or lost to follow-up and each component
- Describe:
  - The correlation of donor chimerism with meeting the primary efficacy endpoint
  - The incidence of donor chimerism (> 50% T cell) and the level of chimerism by visit
  - Change in participant-perceived quality of life
- To evaluate outcomes (de novo donor-specific antibodies [DSA], graft and participant survival, eGFR at 6, 12, and 15 months) in participants who are only transiently chimeric
- To evaluate the incidence of acute rejection (BPAR), death, renal graft loss, and lost to follow-up in participants who did not achieve durable chimerism as well as the ability to wean or remain off IS

Methodology: This is a multicenter, open-label, single-arm proof-of-concept study to demonstrate the safety and efficacy of IF001 in adult participants 3 to 62 months following transplantation of a kidney from a living donor. The same living kidney donor will undergo granulocyte colony-stimulating factor (G-CSF) mobilization and apheresis to generate the IF001 product. Study participants will receive chemotherapy and radiation therapy prior to IF001 cell therapy, then GVHD prophylaxis with high-dose, post-IF001 infusion cyclophosphamide, prednisone, tacrolimus, and mycophenolate mofetil (MMF, or equivalent). Tacrolimus will be withdrawn and discontinued according to weaning criteria.

The study will enroll 15 participants, with no more than 6 who are HLA-matched, to meet a target of 12 participants with complete follow-up.

The sponsor has contracted with the National Marrow Donor Program (NMDP) to manage the donation process and donor follow-up. Donors will undergo stem cell mobilization, stem cell collection, and follow-up according to NMDP standard procedures.

Participant conditioning regimen

Participants will complete their last dose of MMF on Day −10 and last dose of tacrolimus on Day −4 prior to IF001 infusion. Nine days prior to IF001 infusion, participants will begin a reduced intensity conditioning regimen comprising:

- Thymoglobulin® on Days −9 (0.5 mg/kg intravenously [IV] over 6 hours), −8 (2 mg/kg IV over 4 hours), and −7 (2 mg/kg IV over 4 hours); preparations of anti-thymocyte globulin other than Thymoglobulin are excluded
- Fludarabine on Days −6, −5, −4, −3 and −2 (30 mg/m^2 IV daily, dose adjusted for kidney function)
- Cyclophosphamide on Day −6, Day −5 (14.5 mg/kg IV daily), and Days 4 and 5 (50 mg/kg IV daily)
- A single fraction of total body irradiation (TBI) treatment on Day −1 (200 cGY for the HLA-matched cohort and 400 cGy for the partially HLA-matched cohort)

Immunosuppressive therapy

Tacrolimus should be started on Day 6 (24 hours after the completion of post-transplantation cyclophosphamide) and administered as per institutional standards to target a level of 8 to 12 ng/mL.

If tacrolimus is not tolerated, cyclosporine may be substituted IV or orally starting on Day 6 and administered per institutional standards to maintain a level of 200 to 400 ng/mL (high-pressure liquid chromatography method) or 250 to 500 ng/mL (immunoassay method). Recommended starting doses range from 3 to 6 mg/kg/day IV bolus dosing every12 hours or 6 mg/kg/day orally divided twice daily. The calcineurin inhibitor will be continued at therapeutic doses until at minimum Month 6 (Day 183) or longer per institutional preference. Sirolimus with a target concentration of 10 ng/ml may be substituted for participants intolerant of or experiencing unacceptable toxicity from calcineurin inhibitors. The MMF dose will be 15 mg/kg of actual body weight (ABW) orally 3 times daily (TID) up to 1 g TID (or IV equivalent) or an equimolar dose of mycophenolic acid sodium. The MMF dose will be discontinued after the last dose on Day 36 or may continue if active GVHD is present. Prednisone 10 mg orally daily will be initiated on Day 6 with the last dose on Day 36. Filgrastim (G-CSF) or a biosimilar will be given IV or subcutaneously starting on Day 6 for IF001 recipients at 5 µg/kg/day of ABW (rounded to nearest vial size) until the absolute neutrophil count is >1000/µl for 3 days. Additional G-CSF may be administered as warranted. Pegfilgrastim (Neulasta®) and granulocyte-macrophage-CSF are not permitted. Participant IS (tacrolimus or sirolimus or cyclosporine) may be tapered as early as Day 183 provided all the following criteria are met: ● T cell chimerism on Day 160 (±7 days) is >50% ● No active GVHD, and the participant has been off all drugs to treat GVHD (eg, corticosteroids, ruxolitinib) for >30 days without recurrence of GVHD ● No evidence of de novo DSA ● Absence of BPAR of the transplanted kidney In the event of tacrolimus intolerance or based on clinical judgement of the investigator, dose reduction and/or conversion to other primary agents may be necessary and is allowed. Safety monitoring An independent data safety monitoring board (DSMB) will oversee the safety and risk/benefit of the study. The DSMB will follow stopping rules as defined in the DSMB Charter. The DSMB will critically assess the background and clinical course of events leading to the study's temporary halt as soon as possible.
Number of participants (planned): Assuming a 20% dropout rate, it is anticipated that 15 participants will be enrolled, with no more than 6 who are fully HLA-matched, to meet the target of 12 participants with complete follow-up
Diagnosis and main criteria for inclusion: Written informed consent must be obtained from participants before any assessment is performed. Eligible participants will be 18 to 50 years old and must have received a kidney transplant from a partially or fully HLA-matched living donor 3 to 62 months prior to signing informed consent. The participant must be matched with the kidney donor for at least 4 alleles of the following 4 HLA genes: HLA-A, HLA-B, HLA-C, and HLA-DRB1 and at least being a matched allele of the HLA-DRB-1 gene. Participants must have stable renal allograft function ≥ 45 mL/min/1.73 m^2 prior to screening, defined as <25% decrease in eGFR by the CKD-EPI 2021 formula between the last 2 consecutive visits and per investigator judgement. Participants must have a living donor who is 18 to 50 years of age, willing to undergo mobilization and apheresis, and must meet all local standard eligibility criteria to donate stem cells for allogeneic transplantation.

The NMDP criteria for stem cell transplantation must be met for all donors and all participants must meet the clinical site criteria for stem cell transplantation.
Investigational product, dosage, and mode of administration: IF001 is a cell therapy product containing defined doses of CD34+ hematopoietic stem cells and alpha/beta positive T-cell receptor (αβ TCR) cells for induction of transplantation tolerance in recipients of kidney transplants from the same donor. The cryopreserved product contains a minimum of 3 million CD34+ cells per kg of recipient IBW (no maximum) and a target of 10 million αβ TCR cells per kg of recipient IBW (range 8-12 million per kg). The cellular product is infused intravenously through a large bore central venous catheter.
Duration of treatment: IF001 is a one-time infusion administered 3 to 62 months after living donor kidney transplant. Study participation will last approximately 27 months (Screening through Follow-up).
Reference therapy, dosage, and mode of administration: Not applicable.
Criteria for evaluation: Evaluation of all study participants will commence at signing of the informed consent form. The primary and final analysis will occur when all participants have completed the Month 15 visit or have permanently discontinued from the study; cohorts will be analyzed separately. **<u>Primary endpoints</u>:** <u>Safety</u>: ● The incidence and severity of adverse events (AEs; including infections), serious adverse events (SAEs), and AEs leading to study and/or regimen discontinuation <u>Efficacy</u>: ● The proportion of participants who are alive and free from IS, without graft loss or BPAR, at 15 months post-IF001 infusion **<u>Secondary endpoints</u>:** <u>Safety</u>: ● The incidence of and grade of acute GVHD at Day 120 ● The incidence and severity of chronic GVHD at Month 12 ○ All Grade >2 acute GVHD or any chronic GVHD will be reported within 24 hours of site awareness as an adverse event of special interest (AESI) ● Evaluation at Day 183, Month 12, and Month 15: ○ The incidence of BK viremia, viruria, infection, and nephropathy ○ The incidence of AESI according to defined criteria ○ The incidence and grade of acute GVHD ○ The incidence and severity of chronic GVHD ○ Days to neutrophil recovery (>500/µL) ○ Days to platelet recovery (>20K/µL) ○ Number of blood component transfusions ○ The incidence of engraftment syndrome ○ The incidence of primary hematopoietic graft failure ○ The incidence of secondary hematopoietic graft failure

- The incidence and severity of bacterial, fungal, and viral infections

Efficacy:

- **Key Secondary Efficacy Endpoint:** Change in renal function (eGFR by CKD-EPI 2021) from Day −9 (prior to the first dose of transplantation conditioning) to Month 15
- Evaluation at Day 183, Month 12, and Month 15:
  - The incidence of composite endpoint of BPAR, graft loss (renal), or death
  - The incidence of composite endpoint of BPAR, graft loss, death, or lost to follow-up
  - The incidence of graft loss
  - The incidence of death
  - The incidence of lost to follow-up
  - The incidence of acute rejection (BPAR)
  - The incidence of treated BPAR by Banff 2022 criteria
  - The incidence of each type of BPAR by Banff 2022 criteria
  - The incidence of steroid-resistant BPAR
  - The incidence of de novo DSA
  - The incidence of full donor chimerism (>50% T cell), correlated with meeting the primary efficacy endpoint
  - The % chimerism at each visit, correlated with meeting the primary efficacy endpoint
  - Change in renal function (eGFR by CKD-EPI 2021) between Day −9 and Day 183
  - Change in renal function (eGFR by CKD-EPI 2021) between Day −9 and Month 12
  - The incidence or worsening of abnormal histologic findings of cellular or antibody-mediated chronic rejection, chronic glomerulopathy, tubular atrophy and interstitial fibrosis, C4d, calcineurin inhibitor-induced damage, disease recurrence, BK nephropathy between biopsies at Day 160 and Month 15
  - The incidence of renal replacement therapy
  - Participant quality of life according to the 36-Item Short Form Health Survey (SF-36) and End-Stage Renal Disease Symptom Checklist (ESRD-SCL)

Extended evaluation

- Extended Evaluation at Months 18 and 24 (only for events beyond Month 15):
  - The incidence of composite endpoint of BPAR, graft loss (renal), or death
  - The incidence of composite endpoint of BPAR, graft loss, death, or lost to follow-up
  - The incidence of graft loss
  - The incidence of death
  - The incidence of lost to follow-up
  - The incidence of acute rejection (BPAR)
  - The incidence of treated BPAR by Banff 2022 criteria
  - The incidence of each type of BPAR by Banff 2022 criteria
  - The incidence of de novo DSA
  - The incidence of donor chimerism (>50% T cell) and correlation to being off of IS
  - The % chimerism and correlation with remaining off of IS

○ The incidence of acute and chronic GVHD ○ The grade of acute and severity of chronic GVHD
Statistical methods: Statistical methods were not taken into consideration in determining the sample size. The primary efficacy endpoint will be summarized using a point estimate along with its 95% confidence interval (CI) (exact binomial). Participants who are not free from IS at Month 15 or having BPAR, or graft loss, or death will be considered non-responders. Participants who have not died or experienced BPAR or graft loss but are lost to follow-up in the first 15 months after IF001 therapy will be included in the analyses of safety and efficacy. The key secondary efficacy variable is the change in renal function (eGFR by CKD-EPI 2021), from baseline (Day −9, prior to the first dose of transplantation conditioning) to Month 15. Estimated GFR will be derived from local laboratory creatinine values. Because this is a proof-of-concept study, no hypothesis tests will be conducted on the key secondary efficacy endpoint. The mean change in eGFR from Day −9 to Month 15 will be summarized using descriptive statistics and a 95% CI (exact binomial) will be constructed based on the full analysis set. Categorical endpoints will be summarized using counts and proportions. Endpoints with continuous variables will be summarized descriptively (non-missing values, means, standard deviations, medians, 25% and 75% quartiles, minimums, and maximums).

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List of Abbreviations and Definitions of Terms

Abbreviation or Specialist Term	Explanation
αβ TCR	alpha/beta positive T-cell receptor
ABW	actual body weight
AE	adverse event
AESI	adverse event of special interest
ANC	absolute neutrophil count
ATG	anti-thymocyte globulin
BKV	BK virus
BMI	body mass index
BPAR	biopsy-proven acute rejection
CI	confidence interval
CIBMTR	Center for International Blood and Marrow Transplant Research
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
CMV	cytomegalovirus
CRA	Clinical Research Associate
CRF	case report form
CTCAE	Common Terminology Criteria for Adverse Events
DMSO	dimethyl sulfoxide
DSA	donor-specific antibody
DSMB	data safety monitoring board
EAS	Extended Analysis Set
eGFR	estimated glomerular filtration rate
ESRD-SCL	End-Stage Renal Disease Symptom Checklist
FAS	Full Analysis Set
FEV1	forced expiratory volume in one second
FVC	forced vital capacity
GCP	Good Clinical Practice
GVHD	graft-versus-host disease
HCT	hematopoietic cell transplantation
HCV	hepatitis C virus
HIV	human immunodeficiency virus

Abbreviation or Specialist Term	Explanation
HLA	human leukocyte antigen
HSC	hematopoietic stem cells
IBW	ideal body weight
ICF	informed consent form
IRB	Institutional Review Board
IS	immunosuppression
LDKT	living donor kidney transplant
MedDRA	Medical Dictionary for Regulatory Activities
MMF	mycophenolate mofetil
MPS	mycophenolic acid sodium
NMDP	National Marrow Donor Program
PPAS	Per-Protocol Analysis Set
RIC	reduced intensity conditioning
SAE	serious adverse event
SAP	statistical analysis plan
SF-36	36-Item Short Form Health Survey
TBI	total body irradiation
TCR	T-cell receptor
TID	three times daily
WOCBP	women of childbearing potential

This table does not include abbreviations used solely in the tables or figures.

1. INTRODUCTION

1.1. Background

Kidney transplantation is the preferred mode of renal replacement therapy for patients with end-stage kidney failure because it provides better quality of life compared with dialysis and prolongs patient survival. The development of powerful immunosuppressive (IS) medications has reduced the risk of acute rejection and allowed for extended patient and allograft survival. Median graft survival after living donor kidney transplant (LDKT) increased from 12.1 years for transplants in 1995 to 1999 to an estimated 19.2 years for transplants in 2014 to 2017 (Poggio 2021). However, prior kidney transplant failure is now a leading diagnosis for patients on the transplant waiting list as well as a higher proportion of recipients with a previous kidney transplant.

Despite these encouraging trends, chronic IS with standard of care medications imposes a tremendous burden on patients, who require life-long daily medication adherence, experience significant adverse medication effects, and face high medication costs. Adverse events (AEs) related to chronic IS therapy include progression of cardiovascular disease, increased susceptibility to infection, metabolic abnormalities, neurologic disorders, and a greater risk of malignancy. Patients' complex multi-drug regimens require close monitoring and frequent dose adjustments, contributing to incomplete adherence and suboptimal outcomes with respect to renal allograft function, graft survival, and patient survival (Denhaerynck 2005; Fine 2009; FDA 2016; Ettenger 2018).

Beyond the secondary consequences of chronic IS, from the patient perspective, the most important outcome is long-term patient and allograft survival (Husain 2018). This is even more relevant when considering the high mortality rate of patients who return to the waiting list after a failed a kidney transplant (Gill 2002). Conversely, allograft survival is strongly associated with increased patient survival, reduced health care expenditures, and greater likelihood of return to employment.

The development of multiple IS drugs and IS regimens has been key to advances in successful allograft function and survival. Immunosuppressive agents are used for induction (intense IS in the initial days after transplantation), maintenance, and reversal of established rejection (Halloran 2004). Contemporary IS regimens have reduced the incidence of acute rejection in the first post-transplant year to less than 10%, and one-year allograft survival after LDKT is approximately 97% (Lentine 2022). This strong short-term success has not translated to similar long-term success. In fact, late acute cellular rejection, and chronic antibody-mediated rejection (often associated with patient non-adherence to complex IS regimens) is not infrequent and nearly one third of kidney allografts are lost within 10 years (Fine 2009; Ettenger 2018).

Contemporary kidney transplant management includes one or more high-dose IS drugs administered in the hospital at the time of transplant (induction therapy), followed by life-long, daily maintenance treatment (Axelrod 2016). These medications may result in substantial side effects, including nephrotoxicity, neurotoxicity, and metabolic derangements. In addition, these medicines often do not completely control damage to the kidney by the recipients' immune system, ultimately contributing to the kidney's functional decline and, for many, the need for dialysis (graft loss) and retransplant. Therefore, a need for additional options is warranted to address the continued unmet medical need.

1.2. Previous Human Experience

IF001 (previously known as FCR001) is an investigational cell therapy product designed to allow engraftment of allogeneic hematopoietic stem cells (HSCs) across human leukocyte antigen (HLA) mismatch barriers after reduced intensity conditioning (RIC), as evidenced by persistent donor chimerism. IF001 is derived from mobilized peripheral blood cells collected by apheresis from the intended and/or prior kidney donor.

The clinical development program has evaluated the potential for IF001 to induce durable allogeneic tolerance in LDKT recipients to their transplanted organ, thereby eliminating the need for chronic IS after approximately 12 months of a LDKT, without rejecting the transplanted organ.

The current sponsor has changed the alpha/beta positive T-cell receptor (αβ TCR) and CD34+ cell composition of IF001 to optimize donor cell engraftment and reduce the incidence and severity of acute and chronic graft-versus-host disease (GVHD). Please see the IF001 Investigator's Brochure for additional information.

In addition, revisions have been made to recipient eligibility criteria, including prior kidney transplant from a living donor, donor-recipient HLA matching, graft composition, recipient conditioning, and a GVHD prophylaxis regimen (see Section 3.2 and Section 4).

Additional information is provided in the IF001 Investigator's Brochure.

1.2.1. Phase 2 Study FCR001A2201

Study FCR001A2201 was a 2-center, open-label, single-arm trial to investigate whether administration of FCR001 induced donor-specific tolerance to a renal allograft in living donor renal transplant recipients receiving NMA conditioning, and substantially reduce or eliminate the requirement for chronic IS agents within 12 months following transplantation (Leventhal 2015; Leventhal 2018).

Enrollment ended after 37 participants received FCR001. The sponsor terminated the FCR001A2201 study on 23 February 2023 because the LDK transplant clinical development program for FCR001 had been discontinued due to the slow pace of subject enrollment in the FREEDOM-1 Phase 3 clinical study (FCR001A2301), preventing achievement of critical program development milestones in a reasonable timeframe.

The safety profile for study participants was consistent with that expected of the study population (recipient of a kidney transplant and HSC transplant), including all treatments and procedures. There were no events of delayed engraftment or infusion toxicity after FCR001 infusion.

Peripheral blood donor T-cell chimerism 1-month post-transplant ranged between 6-100% and was demonstrated in 35/37 recipients (94.6%) treated with FCR001. Of these, 27 (73%) maintained chimerism >6 months (durable), 8 (21.6%) lost their chimerism (transient), and 2 (5.4%) never engrafted.

Two patients (5.4%) had serious adverse events (SAEs) of GVHD, both with GVHD risk factors (an unrelated female donor to male recipient with low level 1-2/6 HLA matching and cytomegalovirus [CMV] infection).

Three recipients (8.1%) experienced kidney allograft loss during the study; all were transiently chimeric and maintained on chronic IS.

Biopsy-proven allograft rejection occurred only in patients who were either transiently chimeric (6 recipients) or had never engrafted (2 recipients). All biopsies were performed for cause and the patients remained on chronic IS, except for 3 who experienced subsequent graft loss. Of these, 2 had a biopsy-proven acute rejection (BPAR) because of IS discontinuation/marked dose reduction required due to persistent/severe infections and the third graft loss occurred immediately after a BPAR and renal artery thrombosis. The BPAR events in the other 3 patients were successfully treated.

There were 4 deaths in the study, 3 of which were considered by the investigators to be unrelated, with causes of death related to GVHD, lung cancer, sepsis, and respiratory failure.

Twenty-six of the 37 recipients treated with FCR001 (70.3%) successfully discontinued all IS (tacrolimus and mycophenolate mofetil [MMF]), irrespective of donor/recipient relatedness or level of HLA match. As of the cutoff date of the 2023 final report, all 26 maintained persistent chimerism and renal function and had not experienced a BPAR or needed to resume IS therapy during their median 8.9 years (range 3.5-14) of follow up after discontinuing IS.

1.2.2. **Phase 3 Study FCR001A2301**

The Phase 3 study FCR001A2301 (FREEDOM-1, NCT03995901) was planned as a 5-year multicenter, open-label, randomized, SOC-controlled, safety and efficacy of FCR001 (now IF001) in de novo adult LDKT. Recipients were to be randomized 2:1 to the following arms: 1) interventional: IF001 and initial tacrolimus plus MMF, with elimination of IS after 12 months, and 2) control standard of care (induction therapy, tacrolimus, and MMF, with or without corticosteroids).

The primary objective was to evaluate the proportion of recipients free from IS (without BPAR) 24 months post-transplant. The key secondary objective was to evaluate the change in renal function (estimated glomerular filtration rate [eGFR] by Modification of Diet in Renal Disease 4; Levey 2006) from post-transplant baseline (Month 1) to Month 24 in IF001 recipients. The trial was terminated in February 2023 by the previous sponsor for business reasons due to slow enrollment.

As of March 2024, 9 recipient/donor pairs received FCR001 in the FREEDOM-1 clinical study. There was one death due to complications of GVHD, 6 recipients were successfully weaned off IS, one recipient continues to be followed in the IS weaning period, and one recipient continued to receive IS due to ongoing/recurring symptoms of GVHD.

1.3. Risk and Benefit Assessment

The RELIEF trial is a multicenter, open-label, single-arm study to demonstrate the safety and efficacy of IF001 in adult participants 3 to 62 months following transplantation of a kidney from a living donor. The same living kidney donor will undergo mobilization and apheresis to generate the IF001 product. Study participants will receive chemotherapy and radiation therapy prior to IF001 cell therapy, then GVHD prophylaxis with high-dose, post-IF001 infusion cyclophosphamide, prednisone, tacrolimus, and MMF (or equivalent). Tacrolimus will be withdrawn and discontinued according to weaning criteria. The study will enroll 2 cohorts, an HLA- matched cohort (6 participants) and a partially HLA-matched cohort (12 participants). Additional information regarding the design is provided in Section 3.

Living donor renal transplants have been performed for more than 60 years, and renal transplant IS has become the current standard of care. The chronic and daily use of IS agents required for post-transplant renal graft survival, depending on the drug, may increase the risk of opportunistic infection and consequent morbidity and mortality, malignancy (particularly skin and lymphoma), nephrotoxic effects, neurotoxic effects, and long-term toxicities such as diabetes, dyslipidemia, and hypertension. The label of each IS agent provides more detailed information regarding its safety profile.

1.3.1. Risks

1.3.1.1. Conditioning and Stem Cell Transplant/IF001

Conditioning

The key risk associated with RIC (consisting of low-dose TBI, cyclophosphamide, and fludarabine) is cytopenia. Other significant risks associated with RIC are secondary malignancies (TBI, cyclophosphamide), infertility (TBI, cyclophosphamide), infections, and urinary tract and renal toxicity (cyclophosphamide).

Neutropenia/Pancytopenia Prior to Engraftment

In a Phase 2 trial utilizing the RIC used in this study (described in Section 3.2), median time to neutrophil recovery was 17 days (range 14-88 days) and Day 28 cumulative neutrophil recovery was 96% (95% confidence interval [CI] 87-100). Median time to platelet recovery was

25.5 days, with 90% transfusion independence by Day 60. Median time to red blood cell recovery was 25.5 days, with 90% transfusion independence by Day 60 (DeZern 2023).

Bone Marrow Failure

In a Phase 2 trial utilizing the RIC regimen, the proportion of patients alive with engraftment at 1 year was 89% (95% CI 77-100). Twenty-four patients (of 27 total) had sustained >95% donor chimerism in both whole blood and CD3 compartments through 1 year (DeZern 2023).

Graft-Versus-Host Disease

In a Phase 2 study using the RIC regimen, rates of both acute and chronic GVHD were <10% and no patient developed Grade 3 or 4 acute GVHD or moderate/severe chronic GVHD. The cumulative incidence of Grade 2 or 4 acute GVHD on Day 100 was 7%, and the incidence of chronic GVHD at 2 years was 4% (DeZern 2023).

1.3.1.2. **Other Risks**

Renal Graft Loss

Graft loss is a known LDKT complication. The average annual renal graft loss rate for LDKT is 2% per year (USDHHS 2025a).

Malignancy

Malignancy (including premalignant syndromes) is associated with use of IS. Additionally, within a Phase 2 trial using this RIC regimen, 1 out of 27 patients (4%) developed post-transplant lymphoproliferative disorder.

Withdrawal of Immunosuppression

While weaning and withdrawal of IS may increase the risk for acute rejection, none of the recipients withdrawn from IS in the previous Phase 2 study using the investigational product developed acute rejection or chronic changes on histology as of June 2022. In addition, no recipient developed donor-specific antibodies (DSA) and all were reported to have maintained stable renal function 4 to >11 years after withdrawal of IS.

Infection

In a Phase 2 study using the RIC regimen used in this study, 18 patients (67%) experienced infection. Most infections were viral reactivation (CMV [11 infections, with 6 requiring therapy] and Epstein-Barr virus [2 infections, with 1 requiring therapy]). There were also 11 bacterial and 4 fungal infections; none were more severe than Grade 2 (DeZern 2023).

1.3.2. **Benefits**

By establishing durable donor chimerism, the key benefit of IF001 for the renal transplant recipient is to allow freedom from IS and its consequent toxicities while preserving renal function without rejection, even with a completely HLA-mismatched renal allograft. This could represent fulfillment of a long-standing goal in renal transplantation. As freedom from IS can potentially be for life, such patients will benefit from the absence of or significant reduction in side effects of IS, improved long-term outcomes, and improved quality of life.

1.3.3. Risk-Benefit Summary

The clinical data obtained from the earlier Phase 2 FCR001 study supports the use of IF001 for LDKT. A significant efficacy benefit is expected for the recipients with a manageable safety profile in both recipient and donor. Overall, the risks are offset by the potential long-term health benefits to the recipient.

2. Study Objectives

2.1. Primary Objectives

Safety:

- Determine the safety and tolerability of IF001 in previously transplanted recipients of a kidney from a living donor

Efficacy:

- Evaluate the efficacy of IF001 in preventing graft loss and BPAR following removal of pharmacologic IS

2.2. Secondary Objectives

Safety:

- Characterize the incidence and grade of acute GVHD by Day 120 and severity of chronic GVHD by one-year post-IF001 infusion
- Characterize the incidence of BK viremia, viruria, infection, and nephropathy by 15 months post-IF001 infusion
- Characterize the incidence of proteinuria, neurotoxicity, anemia, diabetes, hypertension, dyslipidemia, major adverse cardiovascular events, and malignancies
- Describe:
    - The time from IF001 infusion to neutrophil recovery (>500/µL) and platelet recovery (>20K/µL)
    - The number of blood component transfusions
    - The incidence of engraftment syndrome
    - The incidence of primary and secondary hematopoietic graft failure
    - The incidence and severity of bacterial, fungal, and viral infections

Efficacy:

- Evaluate changes in renal function (eGFR by Chronic Kidney Disease Epidemiology Collaboration 2021 [CKD-EPI] 2021) (Inker 2021) from Day −9, prior to the first dose of transplantation conditioning, to Month 15
- Evaluate the time to the event for the composite of BPAR, graft loss, death, or lost to follow-up and each component
- Describe:
    - The correlation of donor chimerism with meeting the primary efficacy endpoint
    - The incidence of donor chimerism (> 50% T cell) and the level of chimerism by visit
    - Change in participant-perceived quality of life
- To evaluate outcomes (de novo DSA, graft and participant survival, eGFR at 6, 12, and 15 months) in participants who are only transiently chimeric
- To evaluate the incidence of acute rejection (BPAR), death, renal graft loss, and lost to follow-up in participants who did not achieve durable chimerism as well as the ability to wean or remain off IS

Safety and efficacy endpoints are described in Section 8.1.

3. Investigational Plan

3.1. Overall Study Design

Study IF001-CL201 is a Phase 2, multicenter, open-label, single-arm proof-of-concept study to demonstrate the safety and efficacy of IF001 in adult participants 3 to 62 months following transplantation of a kidney from a living donor. The same living kidney donor will undergo G-CSF mobilization and apheresis to generate the IF001 product. The study will be conducted at multiple centers in the United States.

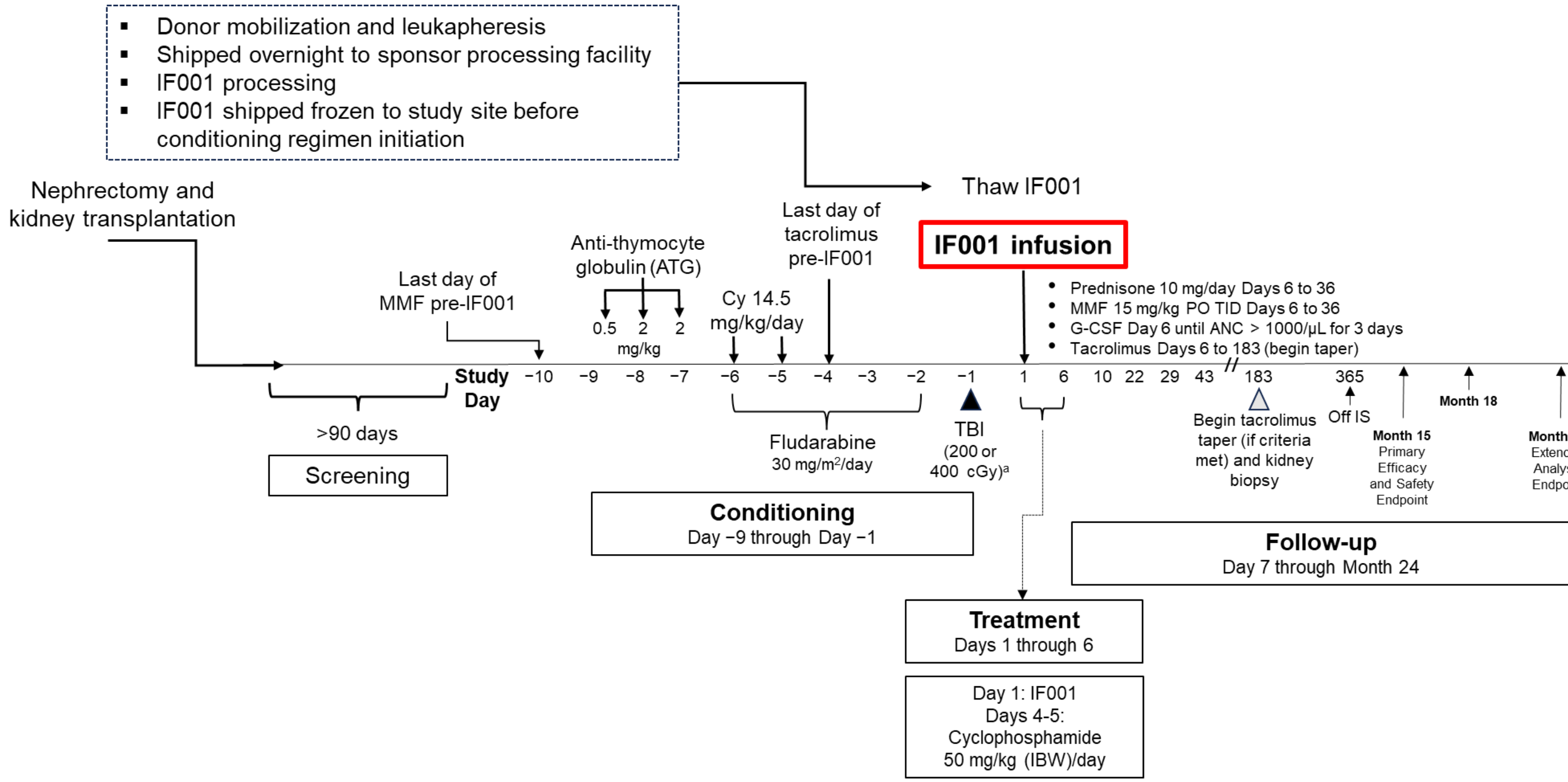
The primary safety endpoint is the incidence and severity of AEs (including infections), SAEs, and AEs leading to study discontinuation and/or discontinuation of immunosuppression wean. The primary efficacy endpoint is the proportion of participants who are alive and free from IS, without graft loss or BPAR, at 15 months post-IF001 infusion.

The study design is based on the trial of DeZern et al. (2023) in patients with aplastic anemia (described in Section 3.2), as it is believed that the knowledge gained from that trial is transferable to kidney transplant recipients. The study will enroll 15 participants, with no more than 6 who are fully HLA-matched, to meet the target of 12 participants with complete follow-up. Each cohort will be analyzed separately.

The study schematic is presented in Figure 1. Screening evaluations will be performed from Day −90 to Day −10. Eligible recipients will receive conditioning for 9 days followed by a 6-day treatment period. Participants will be followed until Month 15 for the primary study period, with an additional Extended Evaluation after Month 15 until Month 24. Study participation will last approximately 27 months (Screening through Follow-up).

All treatment-related AEs and SAEs will be followed until resolution or stabilization. The sponsor has contracted with the National Marrow Donor Program (NMDP) to manage the donation process and donor follow-up.

Figure 1: Study Design



Abbreviations: ANC, absolute neutrophil count; ATG, anti-thymocyte globulin; Cy, cyclophosphamide; G-CSF, granulocyte colony-stimulating factor; HLA, human leukocyte antigen; IBW, ideal body weight; IS, immunosuppression; MMF, mycophenolate mofetil; PO, by mouth; SCT, stem cell transplantation; TID, 3 times daily; TBI, total body irradiation

[a] HLA-matched cohort to receive 200 cGy and partially HLA-matched cohort to receive 400 cGY

3.2. Study Design Rationale

Most of the progress in reducing the toxicity and improving the efficacy of allogeneic hematopoietic cell transplantation (HCT) and in crossing the HLA barrier comes from experience in the treatment of patients with hematologic malignancies and non-malignant hematologic disorders such as aplastic anemia or hemoglobinopathy.

DeZern and colleagues enrolled 27 patients with newly diagnosed aplastic anemia, defined as bone marrow hypocellularity and severe peripheral blood cytopenias. Patients underwent RIC conditioning followed by transplantation of unmanipulated bone marrow from an HLA-haploidentical parent, sibling, child, or second-degree relative. The first 7 patients received a total body irradiation (TBI) dose of 200 cGy; of these, 3 had graft failures and 2 patients died. The dose of TBI was increased to 400 cGy for the next 20 patients, and all 20 are alive and well with >95% donor engraftment in whole blood and CD3 compartments (DeZern 2023). Participants enrolled in the RELIEF trial will be similar those treated by DeZern et al. in that they will not have received extensive prior therapy with T cell-depleting agents or received multiple rounds of IS chemotherapies.

The RELIEF trial is a delayed tolerance protocol. There are 3 key advantages to delaying the HCT for at least 90 days after kidney transplantation:

- The trauma of surgery and ischemia and reperfusion of the transplanted kidney induce inflammation, which can interfere with tolerance induction by hematopoietic stem cell transplantation (Chen 2006; Chong 2012; Braza 2016; Todd 2017)
- Fludarabine used in conditioning for HCT induces severe and potentially lethal neurotoxicity in patients with renal failure (Cheson 1994; Long-Boyle 2011). Correction of renal function by kidney transplantation allows administration of fludarabine without the need for hemodialysis, reducing the potential for significant fludarabine toxicity.
- A delayed tolerance protocol extends the potential benefit of tolerance induction through induction of hematopoietic chimerism to patients who have already had a kidney transplant from a living donor. Expanding the pool of potential patients eligible for tolerance induction allows initial selection of donor-recipient pairs with favorable characteristics for tolerance induction, including close HLA matching.

Concern has been expressed that kidney transplantation is a sensitizing event that prevents subsequent induction of tolerance by HCT from the same donor (Koyama 2007). However, this concern is mitigated by the development of successful protocols for delayed tolerance induction in small and non-human primate animal models (Yamada 2012; Tonsho 2015; Hotta 2018). HCT performed 11 days after kidney transplantation from the same donor achieved mixed chimerism and IS withdrawal in HLA-matched donor-recipient pairs (Busque 2020). One patient received an HLA-matched kidney from his sibling for IgA nephropathy and 14 months later achieved delayed tolerance via HCT from the same donor (Lum 2024).

Additionally, a patient with kidney failure received a kidney from a partially HLA-matched living unrelated donor, developed acute myeloid leukemia, and was successfully transplanted with bone marrow from the same donor 127 days after the kidney transplantation. Six months after bone marrow transplantation the patient successfully discontinued all IS and remained in remission of leukemia with normal renal function at 1-year post-transplant (Yu 2021). In this first reported case of delayed tolerance induction in an HLA-mismatched donor-recipient pair, the conditioning regimen used for transplantation is similar to that of the RELIEF trial.

HLA mismatching between donor and recipient has also been found to be associated with increased incidences of graft failure, GVHD, and non-relapse mortality and decreased overall and event-free survival after related (Anasetti 1989; Anasetti 1990) or unrelated (Lee 2007) donor HCT for hematologic malignancies. To improve patient safety, the RELIEF trial design requires at least 4 allele matches between donor and recipient from among HLA-A, HLA-B, HLA-C, and HLA-DRB1 genes, with at least one of these matches between donor and recipient at HLA-DRB1. This criterion ensures that donor and recipient will have at least 2 HLA Class I allele matches and at least one HLA-DRB1 allele match, so that thymic HLA-restricted CD4 and CD8 T cell responses can be generated after transplantation.

3.3. Dose Rationale

3.3.1. Processed CD34+ Dose

The minimum CD34+ dose is set to be above the lowest dose given in the trial of DeZern et al. (2023), in which all 20 patients conditioned with 400 cGy TBI prior to transplantation achieved engraftment and stable long-term donor chimerism, and were able to discontinue all pharmacologic IS. Among aplastic anemia patients receiving HLA-haploidentical blood or marrow grafts and GVHD prophylaxis incorporating post-transplant cyclophosphamide, the incidence of graft failure was significantly lower for patients receiving grafts with CD34+ cell doses greater than 3.2 million/kg than for patients receiving lower doses of CD34+ cells (11% versus 31%; p=0.02). Thus, higher CD34+ cell doses are associated with faster neutrophil and platelet recovery and a lower risk of graft failure.

This trial has no upper limit to the CD34+ cell dose because increasing CD34 cell doses are associated with faster neutrophil and platelet engraftment (Pedraza 2023) and, when grafts are also depleted of T cells, are not associated with increased incidences of acute or chronic GVHD.

3.3.2. Processed CD3+ T Cell Dose

Dosing of αβ TCR cells in IF001 must consider the risks of acute and chronic GVHD with high αβ TCR cell doses versus the risks of non-engraftment and infection with low αβ TCR cell doses. In the study of DeZern 2023, patients received a median of 5.1 x 10^7 CD3+ cells/kg of recipient ideal body weight (IBW) (range of 2.8 x 10^7 to 8.2 x 10^7). Acute Grade 2 GVHD occurred in 7% (2/27) of patients, and chronic GVHD occurred in 4% (1/27) of patients.

A recent study reported results of a multicenter trial of αβ TCR and CD19-depleted, HLA-haploidentical peripheral blood stem cell transplantation in 30 children and 30 adults with various malignant or non-malignant hematologic disorders (Bethge 2022). Grafts contained a median of 12.1 million CD34+cells/kg (range 4.0 to 54.9 million/kg) and a median of 1.42×10^4 αβ TCR cells/kg (range 0.22 to 6.43 cells/kg). Graft failure occurred in 9/60 patients (15%), and the cumulative incidence of CMV reactivation in the first 100 days was 46% for the entire cohort. These results indicate that near complete removal of T cells bearing the αβ receptor can be associated with increasing incidence of graft failure or infection. The dose of 10 ± 2 million αβ TCR cells/kg in the RELIEF trial was chosen to minimize the incidence of severe acute or chronic GVHD while simultaneously reducing the incidence of graft failure and opportunistic infection.

4. Study Population

4.1. Inclusion Criteria

All participants and donors must fulfill all the following respective criteria. In addition, the NMDP criteria for stem cell transplantation must be met for all donors and all participants must meet the clinical site criteria for stem cell transplantation.

1. Written informed consent must be obtained from participants before any assessment is performed
2. Participant age 18 to 50 years old
3. Donor age 18 to 50 years old
4. Participant must have received a kidney transplant from partially or fully HLA-matched living donor 3 to 62 months prior to signing informed consent
5. The participant must be matched with the kidney donor for at least 4 alleles of the following 4 HLA genes: HLA-A, HLA-B, HLA-C, and HLA-DRB1. The participant must be matched with the kidney donor for at least one allele of the HLA-DRB1 gene.
6. Stable renal allograft function ≥45 mL/min/1.73 m^2 prior to screening (defined as <25% decrease in eGFR by the CKD-EPI 2021 formula [Inker 2021]) between the last 2 consecutive visits and per investigator judgement
7. Must be willing and able to comply with protocol-required visit schedule and visit requirements
8. Currently maintained on tacrolimus and mycophenolate IS

4.2. Exclusion Criteria

1. Participant or donor with use of other investigational drugs within 30 days (or within 5 drug half-lives) of signing informed consent

2. Participant with history of hypersensitivity to any of the medications required for the mobilization or conditioning regimens or to drugs of similar chemical classes
3. Participant and donor who are identical twins
4. Participant who is a pregnant or nursing (lactating) woman
5. Participant with history of malignancy or premalignant syndrome (eg, myelodysplastic syndrome) of any organ system (other than localized excised non-melanomatous lesions of the skin or in-situ cervical cancer), treated or untreated, within the past 5 years, regardless of whether there is evidence of local recurrence or metastases
6. Participant with known bone marrow aplasia
7. History of multi-organ or prior allogeneic hematopoietic progenitor or mesenchymal stem cell transplant prior to signing informed consent
8. Proteinuria >0.5 g/g (500 mg/g) on spot protein/creatinine ratio
9. Participant has presence of DSA (positive DSA defined as a positive crossmatch test of any titer by complement-dependent cytotoxicity or flow cytometric testing or the presence of DSA to HLA-A, B, C, DRB1, DQB1, or DPB1 with mean fluorescence intensity (MFI) >1000 by solid phase immunoassay)
10. History of acute rejection (previously biopsy-proven or suspected and treated) or recurrent kidney disease following living donor kidney transplantation
11. Demonstrated intolerance to maintenance IS with tacrolimus and MMF or mycophenolic acid sodium (MPS)
12. Maintained on oral corticosteroids (prednisone > 10 mg/day or equivalent)
13. History of infection by human immunodeficiency virus (HIV) or hepatitis B virus. Participants with a history of hepatitis C virus (HCV) infection may participate if there is a documented history of treatment with an anti-HCV agent and either one negative polymerase chain reaction (PCR) test at least 3 months after last dose of treatment or 2 negative PCR tests at least 2 weeks apart over a maximum of 4 weeks.
14. Positive for BK virus (BKV), CMV, or Epstein-Barr virus by PCR at screening
15. Active systemic infection or history of recurrent infection (eg, polycystic liver/kidney disease, unless native kidneys removed at time of transplant)
16. Any baseline condition requiring or anticipated will require chronic or intermittent use of systemic steroids or other IS (eg, autoimmune disease, asthma) throughout the course of the study
17. Receipt of live attenuated vaccine administered within 2 months of the start of RIC
18. Body mass index (BMI) <18 or >35 kg/m^2

19. Requiring systemic anticoagulation, (eg, for hyper-coagulation disorders, deep vein thrombosis, atrial fibrillation) that cannot be temporarily interrupted for possible renal biopsy
20. Prior exposure to TBI
21. Forced expiratory volume in one second (FEV1) <50% or forced vital capacity (FVC) <50% of predicted
22. All women of childbearing potential (WOCBP), defined as all women physiologically capable of becoming pregnant, who do not agree to using highly effective methods of contraception during the study. For WOCBP who have completely discontinued mycophenolate, contraception may be discontinued after Month 15 provided no mycophenolate is restarted. Highly effective contraception methods include:
    a. Total abstinence (when this is in line with the preferred and usual lifestyle of the participant). Periodic abstinence (eg, calendar, ovulation, symptothermal, post-ovulation methods) and withdrawal are not acceptable methods of contraception.
    b. Female sterilization (have had surgical bilateral oophorectomy with or without hysterectomy), total hysterectomy, or tubal ligation at least 6 weeks before investigational product infusion. In case of oophorectomy alone, only when the reproductive status of the woman has been confirmed by follow up hormone level assessment.
    c. Male sterilization (at least 6 months prior to screening). For female participants on the study, the vasectomized male partner should be the sole partner.
    d. Use of oral (estrogen and progesterone), injected, or implanted hormonal methods of contraception or placement of an intrauterine device or intrauterine system or other forms of hormonal contraception that have comparable efficacy (failure rate <1%), for example hormone vaginal ring or transdermal hormone contraception.
    e. In case of use of oral contraception women should have been stable on the same medication for a minimum of 3 months before entering study screening
    f. Women are considered post-menopausal and not of childbearing potential if they have had 12 months of natural (spontaneous) amenorrhea with an appropriate clinical profile (eg, age-appropriate, history of vasomotor symptoms) or have had surgical bilateral oophorectomy (with or without hysterectomy) or tubal ligation at least 6 weeks prior to study screening. In the case of oophorectomy alone, only when the reproductive status of the woman has been confirmed by follow up hormone level assessment if she is considered not of childbearing potential.

5. Study Treatments

Donors will undergo stem cell mobilization, stem cell collection, and follow-up according to NMDP standard procedures.

5.1. Description of Investigational Product

IF001 is an investigational cell therapy product containing defined doses of CD34+ HSCs and αβ TCR cells for induction of transplantation tolerance in recipients of kidney transplants from the same donor. The cryopreserved product contains a minimum of 3 million CD34+ cells per kg of recipient IBW (no maximum) and a target of 10 million αβ TCR cells per kg of recipient IBW (range 8-12 million per kg).

5.1.1. Processing, Packaging, and Labeling

Freshly harvested donor stem cells will be immediately transported to the manufacturing center where the IF001 product will be manufactured, cryopreserved, and stored prior to shipping to the study center for thawing and infusion into the participant. The IF001 product will pass applicable release criteria and be shipped frozen to the treating center prior to Day −9 (the initiation of participant's conditioning or related procedures).

Describe packaging for shipment to the clinical site here.

The label will include space to enter the recipient number, content, protocol number, batch number, administration instructions, storage conditions, and cautions. The contents of the label will be in accordance with all applicable regulatory requirements.

5.1.2. Handling and Storage

[placeholder for relevant shipping information, such as temperature monitors used during shipping etc.]. For shipments of IF001 received by the clinical sites, the investigator/designee must confirm the following before distribution and use:

- Appropriate temperature conditions were maintained during transit
- Any discrepancies are documented, reported, and resolved

IF001 should be stored at [storage conditions here] until use. Refer to the IF001 Investigational Product Manual for additional instructions.

Other drugs administered during the study should be handled and stored according to their respective labeling.

5.1.3. Investigational Product Accountability

The investigator is responsible for investigational product accountability, reconciliation, and record maintenance. In accordance with all applicable regulatory requirements, the investigator or designated study center personnel must maintain study investigational product accountability records throughout the course of the study. The amount of investigational product received from the sponsor and the amount infused must be documented. The inventoried supplies will be returned to the sponsor or destroyed on site, after receiving written sponsor approval to do so.

A copy of the final Investigational Product Accountability Log will be provided to the sponsor or designee.

5.2. Participant Numbering

All potential participants must provide informed consent to participate in the study prior to any study assessments or procedures. A unique identifier will be assigned to each participant at the first screening visit.

Patients not meeting the eligibility criteria will be classified as screen failures. Rescreened participants should be recorded as screen failures under their original identification number and assigned a new identification number when consented for re-screening.

5.3. Treatment Blinding

Not applicable as this is an open-label study.

5.4. Infection Prophylaxis

5.4.1. Bacterial Prophylaxis

Bacterial prophylaxis must be provided at least from Day 1 until the absolute neutrophil count (ANC) is >500 cells/µL. Specific antibacterials may be given according to center practice. An example is moxifloxacin 400 mg per day.

5.4.2. Fungal Prophylaxis

Fungal prophylaxis must be provided from Day 1 until the ANC is >500 cells/µL. Specific antifungals may be given according to center practice and investigator judgement but must include coverage of molds such as Aspergillus. An example is posaconazole 300 mg daily.

5.4.3. Viral Prophylaxis

Herpes simplex virus and varicella zoster virus prophylaxis must be provided from Day 1 until Day 183 and is recommended until Month 12. Specific antivirals may be given according to center practice and investigator judgement. An example is valacyclovir 500 mg twice a day.

In addition, for participants who are immunoglobulin G (IgG) (+) for CMV **and** participants who are IgG (−) for CMV but who received a kidney from a donor who is IgG (+), it is recommended that CMV prophylaxis be provided from Day 1 until Day 92. Specific antiviral medications may be given according to center practice and investigator judgement. An example is letermovir 480 mg once daily.

5.4.4. Pneumocystis Prophylaxis

Pneumocystis prophylaxis (Pneumocystis jirovecii pneumonia/pneumocystis carinii pneumonia) must be provided starting on Day 22 or after the ANC is >500 cells/µL, **whichever occurs later**.

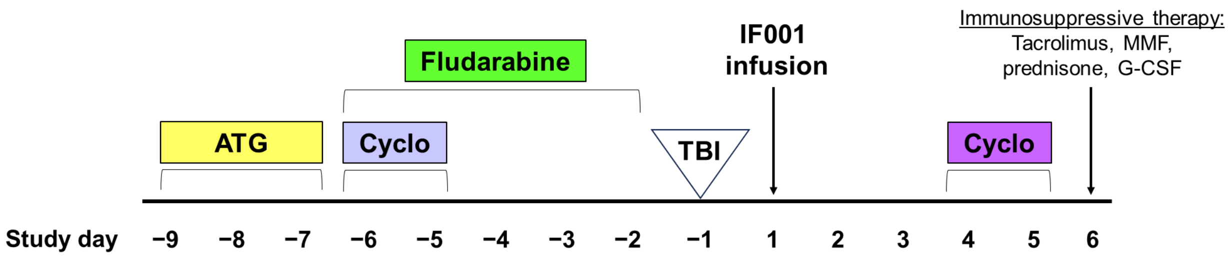
If the CD4 count at the Day 183 visit is >200 cells/µL, the pneumocystis prophylaxis may be discontinued. If the CD4 count is ≤ 200 cells/µL, the agent should be continued until the CD4 count check at Month 12, after which investigator judgement can determine whether to continue or discontinue prophylaxis.

Trimethoprim/sulfamethoxazole 1 single strength tablet daily is the preferred example, with atovaquone 1500 mg once daily or dapsone 100 mg once daily as alternatives.

5.5. Participant Conditioning Regimen

The participant conditioning regimen is illustrated in Figure 2.

Figure 2 Participant Conditioning Regimen



Abbreviations: ATG = anti-thymocyte globulin (Thymoglobulin®); Cyclo = cyclophosphamide; G-CSF = granulocyte colony-stimulating factor; MMF = mycophenolate mofetil; TBI = total body irradiation

Participants will complete their last dose of MMF on Day −10 and last dose of tacrolimus on Day −4 prior to IF001 infusion. Nine days prior to IF001 infusion, participants will begin an RIC regimen:

- Thymoglobulin® Days −9 (0.5 mg/kg intravenously [IV] over 6 hours), −8 (2 mg/kg IV over 4 hours), and −7 (2 mg/kg IV over 4 hours); preparations of anti-thymocyte globulin (ATG) other than Thymoglobulin are excluded
- Fludarabine Days −6, −5, −4, −3 and −2 (30 mg/m^2 IV daily, dose adjusted for kidney function)
    - Fludarabine will be dosed on body surface area based upon the recipient’s actual body weight (ABW) if <125% of IBW. For participants >125% of IBW, fludarabine will be dosed on body surface area based upon adjusted IBW (adjusted IBW = IBW + 0.25[actual – ideal]).
    - For participants who have an estimated or measured creatinine clearance <70 mL/min/1.73 m^2, prior brain radiation, or prior intrathecal chemotherapy, the fludarabine dose should be reduced by 20%. Fludarabine dosing is based on the last creatinine clearance prior to the start of conditioning. Fludarabine dose should be the same on Days −6 to −2, even if the participant’s creatinine changes.
- Cyclophosphamide Day −6, Day −5 (14.5 mg/kg IV daily)
    - Pre-transplantation cyclophosphamide will be administered as a 1- to 2-hour infusion (total dose received 29 mg/kg)

- A single fraction of TBI on Day −1 (200 cGY for the HLA-matched cohort and 400 cGy for the partially HLA-matched cohort)

5.6. IF001 Dosage and Administration

5.6.1. IF001 Infusion

On Day 1, IF001 will be administered intravenously through a large bore central venous catheter. A dialysis-specific catheter/shunt should NOT be used for IF001 administration. See the IF001 Investigational Product Manual for detailed instructions regarding thawing, inspection, and infusion. The thawing procedure, infusion volume, date, and start and stop time will be documented.

The recipient should be carefully monitored during and after infusion of IF001 to detect any changes indicative of hypersensitivity and infusion-related reactions. The infusion will be performed at the recipient's bedside. Premedication with corticosteroids (100 mg hydrocortisone), diphenhydramine, and acetaminophen should be performed as prophylaxis against adverse infusion-related side effects per institutional guidelines.

Cryopreserved products, including IF001, contain dimethyl sulfoxide (DMSO). When the product is thawed and infused, DMSO may cause nausea and/or headache. If reported by the recipient, reduce the rate of infusion and administer ondansetron and/or diphenhydramine (if indicated). Wait 10-15 minutes before thawing and infusing subsequent bags of donor cells.

If a moderate to severe reaction occurs, immediately contact the Medical Monitor.

5.6.1.1. IF001 Dosage Adjustment or Interruption

Once administered, IF001dose adjustments and/or interruptions are not possible as the living cells are in the recipient.

5.6.2. Post-infusion Cyclophosphamide

Cyclophosphamide (50 mg/kg IV daily) will be administered on Day 4 and Day 5 for GVHD prophylaxis. Mesna (sodium 2-mercaptoethane sulfonate) will be administered with cyclophosphamide to minimize potential toxicity of the drug.

Cyclophosphamide and Mesna given post-IF001 infusion will be dosed based on IBW, unless ABW is less than IBW, in which case ABW should be used. If the participant weighs ≥125% IBW, cyclophosphamide will be dosed according to the adjusted IBW.

Participants must be given Mesna 10 mg/kg IBW 30 minutes before and 3, 6, and 8 hours before the Day 4 and Day 5 cyclophosphamide doses.

5.6.3. Immunosuppressive Therapy

Tacrolimus IV or oral (oral immediate-release administration preferred) should be started on Day 6 (at least 24 hours after the completion of post-transplantation cyclophosphamide) and administered as per institutional standards to a target level of 8 to 12 ng/mL. The following considerations should be used when determining a starting dose:

- The recommended IV starting dose is 0.03 mg/kg/day either as a continuous infusion or divided into 12-hour IV bolus dosing based on ABW. Oral equivalent dosing is acceptable at a suggested starting dose of 0.05 to 0.1 mg/kg/dose orally twice daily. If a participant converts from IV to oral once therapeutic levels are achieved, a suggested IV to oral conversion of 1:3 can be used or per institutional standards. Dose adjustment is strongly suggested in participants taking posaconazole or another significant interacting agent.
- Pharmacist assessment of interactions and recommendation for dosing is encouraged

The MMF dose will be 15 mg/kg of ABW orally 3 times daily (TID) up to 1 g TID (or IV equivalent) or an equimolar dose of MPS (see Section 5.6.3.1). The MMF dose will be discontinued after the last dose on Day 36 or may continue if active GVHD is present. Prednisone 10 mg orally daily will be initiated on Day 6 with the last dose on Day 36, unless the participant was on chronic oral steroids prior to enrollment (see Section 5.6.3.1).

Filgrastim (G-CSF) or a biosimilar will be given IV or subcutaneously starting on Day 6 for IF001 recipients at 5 µg/kg/day of ABW (rounded to nearest vial size) until the ANC is >1000 cells/µl for 3 days. Additional G-CSF may be administered as warranted. Pegfilgrastim (Neulasta®) and granulocyte-macrophage-CSF are not permitted.

Participant IS (tacrolimus or sirolimus or cyclosporine) may be tapered as early as Day 183 provided all the following criteria are met and continue to be met until IS weaning is complete:

- T cell chimerism on Day 160 (±7 days) is >50%
- No active GVHD, and the participant has been off all drugs to treat GVHD (eg, corticosteroids, ruxolitinib) for >30 days without recurrence of GVHD
- No evidence of de novo DSA
- Absence of BPAR of the transplanted kidney
    - Note: If a biopsy is medically contraindicated or not feasible, discussion with the Medical Monitor must occur to determine ability to initiate IS weaning

In the event of tacrolimus intolerance or based on clinical judgement of the investigator, dose reduction and/or conversion to other primary agents may be necessary and is allowed.

Table 1: Guidance for Weaning of Tacrolimus

Study visit	Tacrolimus therapeutic range recommendation	Study requirements
Day 6 through Day 183	8 to 12 ng/mL	--
Tapering criteria met to Month 9	4 to 6 ng/mL	Reduce dose by ≥50% by the end of Month 9

Month 9 to Month 12	2 to 4 ng/mL	Discontinue at Month 12 visit
Month 12 to end of study	Off tacrolimus	--

5.6.3.1. Immunosuppression Dosage Adjustment or Interruption

If participants are unable to tolerate the protocol-specified IS dosing scheme, dose adjustments and interruptions are permitted if medically necessary, and should be discussed with the Medical Monitor.

Tacrolimus

If tacrolimus is not tolerated, cyclosporine may be substituted IV or orally starting on Day 6 and administered per institutional standards to maintain a level of 200 to 400 ng/mL (by high-pressure liquid chromatography method) or 250 to 500 ng/mL (immunoassay method). Recommended starting doses range from 3 to 6 mg/kg/day IV bolus dosing every 12 hours or 6 mg/kg/day orally divided twice daily. The calcineurin inhibitor will be continued at therapeutic doses until at a minimum Month 6 (Day 183) or longer per institutional preference. Sirolimus (target concentration of 10 ng/mL) may be substituted for tacrolimus or cyclosporine in case of intolerance of or unacceptable toxicities from calcineurin inhibitors. If there is no evidence of GVHD, then taper as per institutional standard.

MMF

Equimolar doses of enteric-coated MPS (Myfortic® or equivalent) may be substituted for MMF. When switching from MMF immediate-release to MPS delayed-release tablets, the equimolar doses of free mycophenolic acid content are:

- 250 mg MMF = 180 mg MPS
- 500 mg MMF = 360 mg MPS

If neither MMF nor MPS are tolerated, contact the Medical Monitor.

Prednisone or equivalent corticosteroid

If a participant was taking daily corticosteroids for >90 days prior to study enrollment, weaning of corticosteroids to prevent adrenal insufficiency is allowed per institutional standard or clinical judgement, provided that corticosteroids are withdrawn prior to Month 9.

An example of a weaning schedule is provided in Table 2.

Table 2: Example of Corticosteroid Weaning for Participants Previously on Chronic Steroids

Dose	Study days
Prednisone 5mg daily	Days 36 to 57
Prednisone 4mg daily	Days 58 to 92
Prednisone 3mg daily	Days 93 to 120

Prednisone 2mg daily	Days 121 to 160
Prednisone 1.5mg daily	Days 161 to 183
Prednisone 1mg daily	Days 184 to 210 (no visit)
Prednisone 0.5mg daily	Days 211 to 240 (no visit)
Prednisone 0.5mg every other day	Days 241 to Month 9, then discontinue

5.7. Concomitant Medications

The investigator should instruct the participant to notify the study site about any new medications, including non-prescription supplements, he/she takes after study enrollment. All medications, procedures, and significant non-drug therapies (including physical therapy and blood transfusions) administered after the participant has signed the study informed consent form (ICF) will be recorded on the relevant case report form (CRF).

5.7.1. Prohibited Concomitant Medications

The following medications are not permitted during the study:

- Preparations of ATG other than Thymoglobulin
- Pegfilgrastim (Neulasta)
- Granulocyte-macrophage-CSF

5.8. Discontinuation of Treatment and Premature Withdrawal

5.8.1. Discontinuation from Study Treatment

Participants may voluntarily discontinue study treatment or immunosuppression withdrawal for any reason at any time. Such discontinuations should be recorded in the appropriate CRF. The investigator should discontinue other component(s) of the study regimen for a given participant if, on balance, it is believed that continuation would be detrimental to the participant's well-being.

Participants who discontinue study treatment should not be considered withdrawn from the study, and should return for the follow-up assessments as indicated in Table 5. Every effort should be made to contact participants who do not return for follow-up visits (see Section 5.8.3).

5.8.2. Withdrawal of Informed Consent

Participants may voluntarily withdraw their consent to participate in the study for any reason at any time. Withdrawal of consent indicates that the participant does not want to continue in the study, does not want any further visits or assessments, and does not want any further study-related communication. It may also indicate that biologic material that has already been obtained cannot be analyzed.

If a participant withdraws consent, the investigator must make every effort (eg, telephone, email, letter) to determine the primary reason for the decision and record this information. Study participation must be discontinued, and no further assessments conducted. All biological material that has not been analyzed at the time of withdrawal must not be used. Further attempts to contact the participant are not allowed unless safety findings require communication or follow-up.

5.8.3. **Lost to Follow-up**

Participants will be considered lost to follow-up if they meet the following criteria: their status is unclear because of missed study visits without stating the intention to withdraw, they have not responded to repeated communication attempts from the investigator, they did not experience BPAR, graft loss, or death after Day 1 (IF001 infusion), and their last day of contact was prior to Month 15 for the primary analysis and after Month 15 until Month 24 for the extended analysis.

The investigator must demonstrate due diligence to determine the status of the participant. The following actions must be taken if a participant fails to return to the clinic for a required study visit:

- The site must attempt to contact the participant and reschedule the missed visit as soon as possible. In addition, the investigator should counsel the participant on the importance of maintaining the assigned visit schedule and ascertain whether the participant wishes to continue in the study.
- Before a participant is considered lost to follow-up, the investigator or designee must make every effort to re-establish contact with the participant (where possible, 3 telephone calls and, if necessary, a certified letter to the participant's last known mailing address or local equivalent methods). Contact attempts should be documented in the participant's medical record.
- If the participant continues to be unreachable, he/she will be considered to have withdrawn from the study.
- Site personnel will attempt to collect the vital status of the participant within legal and ethical boundaries for all participants enrolled, including those who did not receive IF001. Public sources may be searched for vital status information. If vital status is determined as deceased, this will be documented and the participant will not be considered lost to follow-up. Sponsor personnel will not be involved in any attempts to collect vital status information.

Participants who have not died or experienced BPAR or graft loss but are lost to follow-up in the first 15 months after IF001 therapy will be included in the analyses of safety and efficacy.

5.8.4. Study Completion and Post-study Treatment

Study completion is defined as the point at which all ongoing IF001 recipients have completed the Month 15 visit.

The investigator must provide follow-up medical care for all participants who are prematurely withdrawn from the study or must refer them for appropriate ongoing care according to local practice. This includes reporting to the Center for International Blood and Marrow Transplant Research (CIBMTR) and follow-up in appropriate national registries (eg, United Network for Organ Sharing, etc.).

5.8.5. Enrollment Pause or Early Termination of the Study

The sponsor can pause enrollment or terminate the study at any time for any reason. Should this occur, the sponsor will communicate procedures to be followed to protect the participant's interests. Investigators will be responsible for notifying the Institutional Review Boards (IRBs) of a pause in enrollment or the study termination.

6. Study Procedures and Assessments

Participant study assessments are provided in Table 3, Table 4, and Table 5. Results from assessments will be recorded in the source documents and relevant CRFs.

Patients who signed the informed consent but failed screening will have completed CRFs for the screening period, demographics, screen failure details (giving the reason for screen failure), eligibility criteria, and any SAE data collected. Adverse events that are not SAEs will be followed by the investigator and collected only in the source data.

Re-screening is allowed for patients who were screen failures during the initial screening window, eg, due to laboratory values out of range (see Section 5.2). At the investigator's discretion and following discussion with the Medical Monitor, an extension of screening may be granted under extenuating circumstances such as a death in the family or serious illness.

Visit windows are provided to assist in the scheduling of participants. If a participant misses a visit, it should be rescheduled as close to the original time as possible. It is expected that participants will remain on study for 15 months after IF001 infusion for the primary study analysis and then complete visits at Months 18 and 24 for an extended follow-up analysis of Months >15 through 24.

Table 3: Schedule of Assessments: Screening and Conditioning

Study phase	**Screening**		**Conditioning[1]**								
Time	**Day −90 through −10[2]**		**Day −9 through Day −1**								
Visit (study day)	**Screening**	**−10**	**−9**	**−8**	**−7**	**−6**	**−5**	**−4**	**−3**	**−2**	**−1**
Inclusion/exclusion[3]	X										
Demographics	X										
Medical history	X										
Physical exam	X										
Electrocardiogram	X										
Pulmonary function tests	X										
Height	X										
Vitals including weight[4]	X		X								
Safety panel[5]	X		X	X	X	X	X	X	X	X	X
Urine protein/creatinine ratio	X		X								
Pregnancy test (WOCBP)[6]	X		X								
Vaccine antibody titers[7]	X								X	X	X
Baseline reference chimerism[8]	X										
HLA typing	**X**										
DSA	**X**										
Infectious disease screening[9]	X										
Immunosuppression (MMF)[10]	X	X									

Study phase	**Screening**		**Conditioning**[1]								
Time	**Day −90 through −10**[2]		**Day −9 through Day −1**								
Visit (study day)	**Screening**	**−10**	**−9**	**−8**	**−7**	**−6**	**−5**	**−4**	**−3**	**−2**	**−1**
Immunosuppression (tacrolimus)[10]	X	X	X	X	X	X	X	X			
PRO assessments[11]		X									
Thymoglobulin (ATG)			X	X	X						
Fludarabine						X	X	X	X	X	
Cyclophosphamide 14.5 mg/kg						X	X				
Total body irradiation											X
Adverse events			At each visit								
Concomitant medications	At each visit										

Abbreviations: ATG, anti-thymocyte globulin; DSA, donor-specific antibody; G-CSF, granulocyte colony-stimulating factor; HLA, human leukocyte antigen; MMF, mycophenolate mofetil; NAT, nucleic acid testing; PRO, patient-reported outcome; WOCBP, women of childbearing potential

Note: Assessments with a bold underlined X (**X**) will be run centrally.

1 Before conditioning can begin, adequate reference donor and participant chimerism samples must be available, the participant must have a negative local DSA result, and IF001 must have been received at the clinical site.

2 Donor chimerism baseline reference sample must be confirmed adequate prior to initiating conditioning regimen.

3 Participant eligibility to be verified prior to initiating any conditioning assessments. Latest DSA and cross-match and panel reactive antibodies should be reviewed.

4 Blood pressure will be performed in triplicate (see Section 6.5 for details).

5 Complete blood count with white blood cell differential and comprehensive metabolic panel (see Table 6 for details).

6 Pregnancy tests should be conducted locally (urine or blood) following site standard practice. A urine pregnancy test must be negative for WOCBP before starting filgrastim.

7 Baseline titers to be assessed locally for antibodies, refer to Table 6 for details.

8 At the baseline assessment, 2 aliquots will be drawn in 2 separate ethylenediaminetetraacetic acid tubes. One aliquot will be retained locally and the other will be sent to the central laboratory to reference for post-treatment testing.

9 Refer to Table 6 for details.

10 The last dose of pre-transplantation mycophenolate mofetil is Day −10. The last dose of pre-transplantation tacrolimus is Day −4.

11 Participants will complete the End-Stage Renal Disease Symptom Checklist and the 36-Item Short From Health Survey (SF-36).

Table 4: Schedule of Assessments: Treatment and Initial Follow-Up

Study phase	**Treatment**						**Initial Follow-up**							
Time	**Days 1 through 6**						**Days 7 through 14**							
Visit (study day)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
Safety panel[1]	X	X□	Daily until the ANC is >1000 cells/µL for 3 days											
Urine protein/creatinine ratio	X													
Pregnancy test (WOCBP)[2]	X													
Bacterial/fungal prophylaxis[3]	X	X□	Daily until the ANC >500 cells/µL											
Viral prophylaxis[4]	X	X□	Daily											
IF001 infusion	X													
Cyclophosphamide 50 mg/kg and Mesna 10 mg/kg IBW[5]				X	X									
Filgrastim administration[6]						X□	Daily until the ANC is >1000 cells/µL for 3days							
Immunosuppression (MMF and tacrolimus)[7]						X	X	X	X	X	X	X	X	X
Prednisone						X	X	X	X	X	X	X	X	X
BKV and CMV monitoring[8]								X						
Local tacrolimus level[9]									X□	Daily until ANC >1000 cells/µL for 3 days				
Dialysis	As needed													
Adverse events	At each visit													
Concomitant medications	At each visit													

Abbreviations: ANC, absolute neutrophil count; BKV, BK virus; CMV, cytomegalovirus; G-CSF, granulocyte colony-stimulating factor; GVHD, graft-versus-host disease; IBW, ideal body weight; IgG, immunoglobulin G; IV, intravenously; MMF, mycophenolate mofetil; PCR, polymerase chain reaction; WOCBP, women of childbearing potential

[1] Complete blood count with white blood cell differential and comprehensive metabolic panel (see Table 6 for details). Run locally each day until the ANC is >1000 cells/µL for 3 days.
[2] Pregnancy tests should be conducted locally (urine or blood) following site standard practice. A urine pregnancy test must be negative for women of childbearing potential before starting filgrastim.
[3] Bacterial and fungal prophylaxis must be provided at least from Day 1 until the ANC >500 cells/µL. See Sections 5.4.1 and 5.4.2 for additional information.
[4] Herpes simplex virus and varicella zoster virus prophylaxis must be provided from Day 1 until Day 183 and is recommended until Month 12. In addition, for participants who are IgG (+) for CMV and participants who are IgG (−) for CMV but who received a kidney from a donor who is IgG (+), it is recommended that CMV prophylaxis be provided from Day 1 until Day 92. See Section 5.4.3 for additional information.
[5] Mesna 10 mg/kg IBW to be given 30 minutes before and 3, 6, and 8 hours before the Day 4 and Day 5 cyclophosphamide doses.
[6] Filgrastim (G-CSF) or a biosimilar will be given intravenously or subcutaneously starting on Day 6 at 5 µg/kg/day (rounded to nearest vial size) until the ANC is >1000 cells/µL for 3 days. Additional G-CSF may be administered as warranted. Pegfilgrastim (Neulasta) and GM-CSF are not permitted.
[7] Tacrolimus should be started on Day 6 (at least 24 hours after the completion of post-transplantation cyclophosphamide). The MMF dose will start on Day 6 and may be discontinued after the last dose on Day 36. MMF dosing may continue if active GVHD is present.
[8] BKV and CMV infection monitoring is by local measurement of quantitative viral load by a PCR-based method.
[9] Beginning on Day 9, the recipient should withhold the morning tacrolimus dose until after the blood draw to obtain a trough level. Levels should be assessed daily until the ANC is >1000 cells/µL.

Table 5: Schedule of Assessments: Follow-Up Period

Study phase		**Follow-Up Period**															
Time		**Day 15 through Month 24**															
Visit	**D15**	**D22**	**D29**	**D36**	**D43**	**D50**	**D57**	**D92**	**D120**	**D160**	**D183**	**M9**	**M12**	**M15**	**ET[1]**	**M18[2]**	**M24[2]**
Visit window (+/-)	**2 days**						**3 days**		**7 days**		**14 days**						**21 days**
Physical exam													X		X		
Bacterial/fungal prophylaxis[3]	Daily until the ANC is >500 cells/µL																
Viral prophylaxis[4]	Until Day 183 (until Month 12 recommended)																
Pneumocystis prophylaxis[5]		Start on Day 22 or after the ANC is >500 cells/µL, whichever occurs later															
Vital signs including weight[6]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Safety panel[7]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Urine protein/creatinine ratio	X		X		X		X		X		X		X	X	X		
CD4 count											X		X				
Immunoglobulins											X		X				
Local blood typing[8]													X				
DSA[9]										**X**			**X**	**X**	**X**	**X**	**X**
HAMA sample[10]							**X**										

Study phase		**Follow-Up Period**															
Time		**Day 15 through Month 24**															
Visit	**D15**	**D22**	**D29**	**D36**	**D43**	**D50**	**D57**	**D92**	**D120**	**D160**	**D183**	**M9**	**M12**	**M15**	**ET**[1]	**M18**[2]	**M24**[2]
Visit window (+/-)	**2 days**						**3 days**		**7 days**		**14 days**						**21 days**
Immunosuppression (MMF and tacrolimus)[11]	X	X	X	X	X	X	X	X	X	X	X	X	X	As needed			
Prednisone	X	X	X	X													
Local tacrolimus level[12]	X	X	X	X	X	X	X	X	X	X	X	X	X	As needed			
Filgrastim administration[13]	Daily until the ANC is >1000 cells/µL for 3days							As needed									
BKV and CMV monitoring[14]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
GVHD assessment[15]		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X
Kidney allograft biopsy[16]				As needed						**X**		As needed		**X**	As needed		
Donor chimerism[17]			X				X			**X**			**X**	**X**	**X**	**X**	**X**
PRO assessments[18]													X		X		
Adverse events	At each visit																
Concomitant medications	At each visit																

Abbreviations: BKV, BK virus; CMV, cytomegalovirus; D, day; DSA, donor-specific antibody; ET, early termination; G-CSF, granulocyte colony-stimulating factor; GVHD, graft-versus-host disease; HAMA, human anti-mouse antibody; M, month; MMF, mycophenolate mofetil; PRO, patient-reported outcomes; WOCBP, women of childbearing potential

Note: Assessments with a bold underlined X (**<u>X</u>**) will be run centrally.

[1] Early termination procedures will be conducted in participants who prematurely discontinue from the study before Month 15. See Section 5.8 for additional information.

[2] Visits at Month 18 and Month 24 can be conducted virtually for participants who cannot return to the study center. In these cases safety laboratory samples will be run centrally.

[3] Bacterial and fungal prophylaxis must be provided at least from Day 1 until the ANC >500 cells/µL. See Sections 5.4.1 and 5.4.2 for additional information.

[4] Herpes simplex virus and varicella zoster virus prophylaxis must be provided from Day 1 until Day 183 and is recommended until Month 12. In addition, for participants who are IgG (+) for CMV and participants who are IgG (−) for CMV but who received a kidney from a donor who is IgG (+), it is recommended that CMV prophylaxis be provided from Day 1 until Day 92. See Section 5.4.3 for additional information.

[5] Pneumocystis prophylaxis (Pneumocystis jirovecii pneumonia/pneumocystis carinii pneumonia) must be provided starting on Day 22 or after the ANC is >500 cells/µL, **<u>whichever occurs later</u>**. Additional information is provided in Section 5.4.4.

[6] Blood pressure will be performed in triplicate (see Section 6.5 for details).

[7] Complete blood count with white blood cell differential and comprehensive metabolic panel (see Table 6 for details).

[8] ABO/Rh blood typing should be performed locally at Month 12 post-IF001 infusion and whenever (prior) a blood transfusion is given.

[9] Specimens for DSA should also be drawn and tested locally at the time of any suspected rejection.

[10] A HAMA sample will be collected and analyzed centrally on serum.

[11] After complete weaning of immunosuppression additional serum creatinine should be performed locally weekly for 4 weeks.

[12] The recipient should withhold the morning tacrolimus dose until after the blood draw to obtain a trough level. Levels should be assessed daily until the ANC is >1000 cells/µL. Thereafter tacrolimus levels should be assessed 3 days after any dose adjustment.

[13] Filgrastim (G-CSF) or a biosimilar given intravenously or subcutaneously 5 µg/kg/day (rounded to nearest vial size) should be administered until the ANC is >1000 cells/µL for 3 days. Additional G-CSF may be administered as warranted. Pegfilgrastim (Neulasta) and GM-CSF are not permitted.

[14] BKV and CMV infection monitoring is by local measurement of quantitative viral load by a PCR-based method. <u>BK and CMV monitoring are performed more frequently than scheduled visits displayed in this table; s</u>ee Section 6.9 for details.

[15] Beginning at Day 22 through Day 92, only acute GVHD assessments will be performed. At Days 92 and 120, both acute and chronic GVHD assessments will be performed. Beyond Day 120, chronic GVHD assessments will be performed routinely, but acute GVHD also can be assessed as clinically warranted. If acute or chronic GVHD is suspected or detected at any time between visits, regardless of location or grade, appropriate GVHD assessment must be performed at an unscheduled visit. Sites should supplement with weekly telephone contact through Day 183. All Grade ≥2 acute GVHD or moderate to severe chronic GVHD must be reported within 24 hours of site awareness. See Section 6.11 for details.

[16] Kidney allograft protocol biopsies are to be performed and analyzed centrally at Day 160 and at Month 15; see Section 6.16. Site-specific protocol/surveillance biopsies without suspicion of acute rejection are prohibited. Additional information regarding acute rejection is provided in Section 6.13.

[17] Whole blood and T cell chimerism require a whole blood sample.

[18] Participants will complete the End-Stage Renal Disease Symptom Checklist and the 36-Item Short Form Health Survey (SF-36).

6.1. Obtaining Informed Consent

Signed informed consent will be obtained from the participant before any study procedures are undertaken. Details regarding the informed consent process and documentation are provided in Section 13.2.

6.2. Participant Demographics and Other Baseline Characteristics

Participant and other baseline demographic characteristics are described in Section 8.5.

6.3. Medical History

Medical history findings (ie, previous diagnoses, diseases, or surgeries) not pertaining to the study indication, started before signing the informed consent and during the screening phase, and considered relevant for the participant's study eligibility will be collected and captured in the CRF.

6.4. Physical Examination

Physical examinations will include Karnofsky performance status (Péus 2013), general appearance, head/eyes/ears/nose/throat, lungs, cardiac, abdominal, skin, extremities, lymph nodes, and a neurologic examination. In addition to the protocol-specified times, physical assessments can be made at the investigator's discretion to evaluate symptoms or AEs.

6.5. Vital Signs Including Height and Weight

Vital sign assessments (blood pressure and pulse rate) should be performed by the same study site staff member using the same validated device; if a different device is used, please note in source documents.

Blood pressure and pulse rate will be assessed in the same arm each time of determination and after the participant has rested in the sitting position (may be supine if during hospitalization) for at least 5 minutes. The person taking the blood pressure measurement should check whether the participant is calm before taking the reading. Systolic and diastolic blood pressure will be measured using an automated validated device with an appropriately sized cuff. The sitting measurements will be repeated at 1- to 2-minute intervals for a total of 3 measurements. If available cuff sizes are not large enough for the participant's arm circumference, a sphygmomanometer with an appropriately sized cuff may be used.

Participants should be weighed without shoes or coat using the same scale. Height (without shoes) will be measured only at the first screening visit. The BMI will be calculated at each visit using current weight and height at the first screening visit.

6.6. Electrocardiogram

A 12-lead electrocardiogram will be conducted at screening for all participants.

6.7. Pulmonary Function Tests

Pulmonary function tests including FEV1 and FVC will be conducted at screening for all participants.

6.8. Revaccination

The recommendations for vaccination after allogeneic HCT are based on the guidelines of the Infectious Disease Society of America (Rubin 2014) and are summarized in Appendix 1.

6.9. Laboratory Assessments

Local and central laboratories will be used for analysis of specimens collected as described in the schedules of assessments (Table 3, Table 4, and Table 5). Clinical laboratory tests can also be performed as medically indicated. Details regarding collections, shipment of samples, and results reporting will be provided in the Laboratory Manual. Clinical laboratory assessments are summarized in Table 6.

BKV and CMV infection monitoring is by local measurement of quantitative viral load by a PCR-based method. BK virus and CMV monitoring are performed more frequently than scheduled visits displayed in the schedules of assessments: Weekly through Day 183 (Month 6) (± 1 day through Day 7, ± 2 days through Day 50, ± 3 days through Day 92, and ± 1-2 weeks [per schedule of assessments] through Day 183 [Month 6]), every 2 weeks (± 14 days) thereafter through Month 12, and every 3 months (± 21 days) through Month 15.

All laboratory test results considered clinically significant by the investigator will be followed to a satisfactory resolution. A laboratory test value that requires treatment or requires a participant to be discontinued from treatment or the study will be recorded as an AE (see Section 7.1.3).

Table 6 Clinical Laboratory Assessments

Complete blood count	**Comprehensive metabolic panel**
White blood cell count	Albumin
Red blood cell count	Alanine aminotransferase
Hemoglobin	Aspartate aminotransferase
Hematocrit	Alkaline phosphatase
Mean corpuscular volume	Total bilirubin
Mean corpuscular hemoglobin	Blood urea nitrogen
Mean corpuscular hemoglobin concentration	Creatinine
Mean platelet volume	Calcium
Red blood cell distribution width	Chloride
Platelet count	Sodium
	Potassium
Differential (electronic)	Glucose
Neutrophils %	Total protein
Immature granulocytes %	Carbon dioxide
Lymphocytes %	
Monocytes %	**Infectious disease screening**
Eosinophils %	PCR:
Basophils %	BKV
Absolute neutrophil count	CMV
Immature granulocytes, absolute	EBV
Lymphocytes, absolute	Hepatitis C virus
Monocytes, absolute	HIV
Eosinophils, absolute	Hepatitis B virus testing:
Nucleated red blood cell number	Hepatitis B core antibody
Platelet estimate	Hepatitis B surface antibody
	Hepatitis B surface antigen
Pregnancy	
Serum or urine	**Viral load monitoring (PCR)**
	BKV
Vaccine antibody titers	CMV
Varicella	
Measles	**Urinalysis**
Mumps	Protein/creatinine ratio
Rubella	
Hepatitis B virus	**Other**
Tetanus	HLA typing
Diphtheria	Donor-specific antibody
Meningococcus	ABO blood typing and Rh factor
Pneumococcus	Donor whole blood and blood T cell chimerism
	Tacrolimus (trough)
Immunoglobulins	HAMA sample
IgA	
IgG	
IgM	

Abbreviations: BKV, BK virus; CMV, cytomegalovirus; EBV, Epstein-Barr virus; HAMA, human anti-mouse antibody; HIV, human immunodeficiency virus; HLA, human leukocyte antigen; IgA, immunoglobulin A; IgG, immunoglobulin G; IgM, immunoglobulin M; PCR, polymerase chain reaction

6.10. Renal Function

Renal function by estimated eGFR using the CKD-EPI 2021 formula will be derived from local laboratory creatinine values. This will also be the primary method of assessment of renal function with respect to other renal endpoints.

6.11. Graft-versus-Host-Disease

GVHD is considered an adverse event of special interest (AESI; see Section 7.4.3). Beginning at Day 22 through Day 92, only acute GVHD assessments will be performed. At Day 92 and Day 120, both acute and chronic GVHD assessments will be performed. Beyond Day 120, chronic GVHD assessments will be performed routinely, but acute GVHD also can be assessed as clinically warranted.

If acute or chronic GVHD is suspected or detected at any time between visits, regardless of location or grade, appropriate GVHD assessment must be performed at an unscheduled visit. Sites should supplement with weekly telephone contact through Day 183.

Acute and chronic GVHD will be managed according to institutional practices. Grading of GVHD should follow the criteria provided in Appendix 2 for acute GVHD (Harris 2016) and Appendix 3 for chronic GVHD (Jagasia 2015).

6.12. Off Immunosuppression

A participant will be considered free from IS if they are not currently taking any IS agents and have not taken IS since withdrawal. An exception will be brief use of systemic corticosteroids for no more than 8 weeks (total duration) if administered for an indication other than rejection prophylaxis or treatment (such as newly developed or recurrent autoimmune diseases, asthma, hypersensitivity reactions). The date of discontinuation of the last IS agent will be captured on the relevant CRF.

6.13. Acute Rejection

A BPAR is defined as a biopsy result with either T cell-mediated rejection of Banff Grade ≥1A OR acute/active antibody-mediated rejection (local pathology reading according to the Banff 2022 criteria (Appendix 4; Naesens 2024). Renal biopsies will be performed for all cases of suspected acute rejection and at specified protocol visits (see Section 6.16). Site-specific protocol/surveillance biopsies without suspicion of acute rejection are **<u>prohibited</u>**.

Suspicion of acute rejection is deferred to the clinical investigator and local practices. An example would be: an acute and persistent increase in serum creatinine by ≥20% and/or new onset or worsening proteinuria, with other potential causes ruled out. <u>Clinicians must discuss findings with the Medical Monitor prior to proceeding to biopsy</u>.

All episodes of acute rejection must be entered on the relevant CRF(s) preferably within 24 hours. Acute rejections should not generally be considered an AE unless unusually severe in presentation.

Acute rejections should be treated with bolus methylprednisolone (other corticosteroids are acceptable at an equivalent dose) according to local practice. Recommended treatment is a total dose of at least 750 mg methylprednisolone divided into 2 or 3 once-daily IV boluses. Other anti-rejection therapies (ie, T cell-depleting antibody therapy) should only be used in cases of steroid-resistant rejections, vascular rejections, antibody-mediated rejections, or rejections with a Banff Grade ≥ 2B.

All medications used for the treatment of suspected or confirmed acute rejections must be recorded on the relevant CRF.

6.14. **Graft Loss**

The allograft will be presumed to be lost on the day the participant starts dialysis and is subsequently unable to be removed from dialysis for 56 consecutive days or if the participant is retransplanted. If the participant undergoes allograft nephrectomy prior to starting permanent dialysis, then the day of nephrectomy is the day of graft loss. The reason for graft loss and study discontinuation will be recorded on the relevant CRFs. Graft loss is considered an SAE and should be reported as described in Section 7.4.2.

6.15. **Death**

In the event of a participant's death, the SAE leading to death should be reported to the sponsor or designee as described in Section 7.4.2 and Section 11.2.2.

6.16. **Renal Allograft Biopsy**

Participants who cannot have protocol-specified biopsies for medical reasons should be discussed with the Medical Monitor.

Biopsies will be read by the local pathologist according to the updated Banff 2022 criteria (Appendix 4; Naesens 2024) and the results of the recorded on the relevant CRF. The local results will be used for the management of acute rejection; see Section 6.13 for additional information.

Central pathology diagnosis will be used for analysis of efficacy endpoints.

6.17. **Patient-Reported Outcome Measures**

The impact of IF001 on the participants' quality of life will be assessed by the 36-Item Short Form Health Survey (SF-36; RAND 2025, Appendix 5) and the End-Stage Renal Disease Symptom Checklist (ESRD-SCL; Franke 2016, Appendix 6). The SF-36 is a widely used tool and assesses overall physical function, general health, and social/psychological factors and has been validated in the end-stage renal disease population. The ESRD-SCL was developed specifically for and validated in the end-stage renal disease population.

The questionnaires should be completed by participants at the clinic before seeing the study physician. Participants should be given sufficient instruction, space, time, and privacy to complete the questionnaire. Study staff should check the questionnaire responses for completeness and encourage the participant to provide any missing responses. The responses from the paper questionnaire will be entered into the CRF by study staff.

Before the clinical examination, completed questionnaires will be reviewed by the investigator for responses that might indicate possible AEs or SAEs. The investigator should review the responses to the questions as well as any unsolicited comments recorded by the participant. Confirmed AEs or SAEs must be recorded as described in Section 7.

6.18. Appropriateness of Measures

All assessments used in this study are widely used and generally recognized as reliable, accurate, and relevant.

The efficacy endpoints of BPAR, graft loss, death, or loss to follow-up have all been used as endpoints in many previous studies in the kidney transplantation indication and are widely accepted by health authorities for global registration purposes in this indication. The endpoint of freedom from IS is the target of this therapy, without which efficacy cannot be adequately measured. In addition, eGFR has been utilized in registration studies as a key secondary or supportive endpoint as a measure of transplanted organ function.

7. Adverse Events

7.1. Definition of an Adverse Event

An AE is any untoward medical occurrence (ie, any unfavorable and unintended sign [including abnormal laboratory findings], symptom, or disease) in a participant or clinical investigation participant after providing written informed consent for participation in the study. Therefore, an AE may or may not be temporally or causally associated with the use of a medicinal (investigational) product.

Investigators are responsible for managing the safety of participants and identifying AEs. The occurrence of AEs should be sought by non-directive questioning of the participant at each study visit. Adverse events also may be detected when volunteered by the participant during or between visits or through physical examination (or body system assessment), laboratory test, or other assessments.

Guidance for the investigator regarding safety and common side effects is provided in the IF001 Investigator's Brochure.

7.1.1. Assessment of Severity

Severity will be assessed per the Common Terminology Criteria for Adverse Events (CTCAE) v 6.0 (USDHHS 2025b) Grade 1 through Grade 5. If an AE is not specifically graded by CTCAE v6.0, the investigator should use medical judgement to determine AE severity following the general CTCAE v6.0 guidelines of Grade 1 = mild, Grade 2 = moderate, Grade 3 = severe, Grade 4 = life-threatening, or Grade 5 = death.

7.1.2. Assessment of Causality

The investigator must assess the relationship of each AE/SAE with the use of the investigational product as defined in Section 5.6 using their medical judgement. The following definitions will be used:

Related: There is a reasonable possibility that the event is related to the investigational product IF001 (eg, there is a temporal association between the treatment exposure and onset of the AE, the manifestations of the AE are consistent with the known product profile within the treatment, the event cannot be reasonably explained by alternative causes).

The investigator should use their best clinical judgement to determine if the AE is at least possibly related to IF001. If so, this assessment will be considered "related."

Separately, the investigator will use their best clinical judgement to assess whether the AE is associated with other study interventions such as the RIC regimen and/or IS regimen. Although the association will be captured, these events will not be considered "related" from a regulatory standpoint since the investigator does not consider them related to the IF001 investigational product.

Not related: The AE is related to a concomitant medical condition, concomitant medication(s), or an accident, with no reasonable association with the investigational product IF001.

7.1.3. Laboratory Test Abnormalities

Abnormal laboratory values or test results constitute AEs only if they fulfill at least one of the following criteria:

- They induce clinical signs or symptoms
- They are considered clinically significant
- They require therapy

Clinically significant abnormal laboratory values or test results should be identified through a review of values outside of normal ranges/clinically notable ranges, significant changes from baseline or the previous visit, or values considered to be non-typical in participant with underlying disease. Liver toxicities should be graded according to CTCAE v6.0.

7.2. Definition of a Serious Adverse Event

An SAE is defined as any AE appearance of (or worsening of any pre-existing) undesirable sign(s), symptom(s), or medical conditions(s) which meets any one of the following criteria:

- Is fatal
- Life-threatening
- Results in persistent or significant disability/incapacity
- Constitutes a congenital anomaly/birth defect
- Requires inpatient hospitalization or prolongation of existing hospitalization, unless hospitalization is solely for:
  - Renal allograft biopsy
  - Treatment of acute rejection
  - Routine treatment or monitoring of the studied indication, not associated with any deterioration in condition
  - Elective or pre-planned treatment for a pre-existing condition that is unrelated to the indication under study and has not worsened since signing the informed consent
  - Treatment on an emergency outpatient basis for an event not fulfilling any of the definitions of an SAE given above and not resulting in hospital admission
  - Social reasons and respite care in the absence of any deterioration in the participant's general condition
- Is medically significant (ie, defined as an event that jeopardizes the participant or may require medical or surgical intervention)

All malignant neoplasms will be assessed as serious under "medically significant" if other seriousness criteria are not met.

Hematologic AEs should not be reported as SAEs unless they occur 4 or more weeks after IF001 infusion.

Life-threatening in the context of an SAE refers to an event in which the participant was at risk of death at the time of the event; it does not refer to an event that hypothetically might have caused death if more severe.

Medical and scientific judgement should be exercised in deciding whether other situations should be considered SAEs, such as important medical events that might not be immediately life-threatening or result in death or hospitalization but might jeopardize the participant or might require intervention to prevent one of the other outcomes listed above. Examples of such events are intensive treatment in an emergency room or at home for allergic bronchospasm, blood dyscrasias, or convulsions that do not result in hospitalization or development of dependency or abuse.

Any suspected transmission via a medicinal product of an infectious agent is also considered a serious adverse reaction.

All SAEs will be reported following the instructions provided in Section 7.4.2.

7.3. Following Adverse Events

Once an AE is detected, it should be followed until its resolution or until it is judged to be permanent. An assessment should be made at each visit (or more frequently, if necessary) of any changes in severity, the suspected relationship to the investigational product, the interventions required to treat it, and the outcome.

7.4. Reporting Adverse Events

7.4.1. Adverse Event Reporting Period

After the informed consent has been signed, all AEs and SAEs, regardless of relationship to IF001, will be reported through completion of the last study visit of the primary analysis period (Month 15). After this period, the investigator should report any SAEs that are believed to be related to IF001 (see Section 7.4.5).

7.4.2. Reporting Serious Adverse Events

Every SAE, regardless of causality, occurring after the participant has provided informed consent and through completion of the last study visit must be reported to the sponsor or designee within 24 hours of learning of its occurrence. Any SAEs experienced after the last study visit should only be reported to the sponsor or designee if the investigator suspects a causal relationship to IF001 (see Section 7.4.5).

Recurrent symptoms of an overlying reported SAE, complications, or progression of the initial SAE must be reported as follow-up to the original episode, regardless of when the event occurs. This report must be submitted within 24 hours of the investigator receiving the follow-up information. An SAE that is considered completely unrelated to a previously reported one should be reported separately as a new event.

All SAEs must be reported to the sponsor or designee within 24 hours of study personnel's first awareness of the event by filling out the corresponding CRF page. The investigator must be aware of the SAE reported and verify the accuracy of the information recorded on the CRF with the corresponding source documents. See Section 11.2.2 for instructions for reporting recipient deaths.

If there are network outages, a paper copy of the AE form must be completed and emailed to Emmes (contracted by the sponsor) to initiate sponsor review. Once the system is available, the event information must be entered into the system as soon as possible.

Additional follow-up information, if required or available, should be communicated to the sponsor or designee within one business day of receipt.

It is the Principal Investigator's responsibility to notify the IRB of all SAEs that occur at his or her site.

7.4.3. Adverse Events of Special Interest

All Grade ≥2 acute GVHD or moderate to severe chronic GVHD must be reported within 24 hours of site awareness. Other AESI will be detailed in the statistical analysis plan (SAP).

7.4.4. Pregnancy

Any pregnancy must be reported to the sponsor and/or its designee within 24 hours of learning of its occurrence during the study. The pregnancy should be followed to determine outcome, including spontaneous or voluntary termination, details of the birth, and the presence or absence of any birth defects, congenital abnormalities, or maternal and/or newborn complications.

7.4.5. Post-Study Adverse Events

The investigator should instruct each participant to report any new AE beyond the protocol observation period that the participant or their personal physician believes might reasonably be related to IF001. This information should be recorded in the investigator's source documents, however, if the AE meets the criteria of an SAE, it must be reported to the sponsor.

7.4.6. Expedited Reporting

For determination of whether an SAE requires expedited regulatory reporting to relevant health authorities, an AE is considered "unexpected" if it is not listed in the Reference Safety Information section of the current IF001 Investigator's Brochure.

The sponsor is responsible for notifying the relevant regulatory authorities and Data Safety Monitoring Board (DSMB) of certain events. Investigators will also be notified of all unexpected, serious, investigational product-related events (7-day or 15-day Safety Reports) that occur during the study. Each Principal Investigator is responsible for notifying the IRB of these additional SAEs.

8. Statistical Considerations and Analytical Plan

8.1. Study Endpoints

8.1.1. Primary Endpoints

The primary safety endpoint is the incidence and severity of AEs (including infections), SAEs, and AEs leading to study and/or regimen discontinuation.

The primary efficacy endpoint is the proportion of participants who are alive and free from IS, without graft loss or BPAR, at 15 months post-IF001 infusion.

8.1.2. Secondary Endpoints

Safety

- The incidence of and grade of acute GVHD at Day 120
- The incidence and severity of chronic GVHD at Month 12
    - o All Grade >2 acute GVHD or any chronic GVHD will be reported within 24 hours of site awareness as an AESI
- Evaluation at Day 183, Month 12, and Month 15:
    - The incidence of BK viremia, viruria, infection, and nephropathy
    - The incidence of AESI according to defined criteria
    - The incidence and grade of acute GVHD
    - The incidence and severity of chronic GVHD
    - Days to neutrophil recovery (>500/µL)
    - Days to platelet recovery (>20K/µL)
    - Number of blood component transfusions
    - The incidence of engraftment syndrome
    - The incidence of primary hematopoietic graft failure
    - The incidence of secondary hematopoietic graft failure
    - The incidence and severity of bacterial, fungal, and viral infections

Efficacy

- **Key Secondary Efficacy Endpoint**: Change in renal function (eGFR by CKD-EPI 2021; [Inker 2021]) from Day −9 to Month 15
- Evaluation at Day 183, Month 12, and Month 15:
    - The incidence of composite endpoint of BPAR, graft loss (renal), or death
    - The incidence of composite endpoint of BPAR, graft loss, death. or lost to follow-up
    - The incidence of graft loss
    - The incidence of death
    - The incidence of lost to follow-up
    - The incidence of acute rejection (BPAR)
    - The incidence of treated BPAR by Banff 2022 criteria (Naesens 2024)

- The incidence of each type of BPAR by Banff 2022 criteria
- The incidence of steroid-resistant BPAR
- The incidence of de novo DSA
- The incidence of full donor chimerism (>50% T cell), correlated with meeting the primary efficacy endpoint
- The % chimerism at each visit, correlated with meeting the primary efficacy endpoint
- Change in renal function (eGFR by CKD-EPI 2021) between Day −9 and Day 183
- Change in renal function (eGFR by CKD-EPI 2021) between Day −9 and Month 12
- The incidence or worsening of abnormal histologic findings of cellular or antibody-mediated chronic rejection, chronic glomerulopathy, tubular atrophy and interstitial fibrosis, C4d, calcineurin inhibitor-induced damage, disease recurrence, BK nephropathy between biopsies at Day 160 and Month 15
- The incidence of renal replacement therapy
- Participant quality of life according to the SF-36 (RAND 2025) and ESRD-SCL (Franke 2016)

Extended evaluation

- Extended Evaluation at Months 18 and 24 (only for events beyond Month 15):
  - The incidence of composite endpoint of BPAR, graft loss (renal), or death
  - The incidence of composite endpoint of BPAR, graft loss, death, or lost to follow-up
  - The incidence of graft loss
  - The incidence of death
  - The incidence of lost to follow-up
  - The incidence of acute rejection (BPAR)
  - The incidence of treated BPAR by Banff 2022 criteria
  - The incidence of each type of BPAR by Banff 2022 criteria
  - The incidence of de novo DSA
  - The incidence of donor chimerism (>50% T cell) and correlation to being off of IS
  - The % chimerism and correlation with remaining off of IS
  - The incidence of acute and chronic GVHD
  - The grade of acute and severity of chronic GVHD

8.2. General Considerations

The SAP will be finalized and approved prior to the database lock and will include a more technical and detailed description of the statistical analyses described in this section. Any deviations from the planned statistical analysis will be summarized in the clinical study report.

All descriptive statistical analyses will be performed using SAS version 9.4 or higher. For categorical variables, the number and percentage within each category of the parameter will be calculated (including a category labeled as "missing" when appropriate). For continuous variables, the number of participants with no missing data (n), mean, median, 25% and 75% quartiles, standard deviation (SD), minimum, and maximum values will be presented.

Baseline is defined as the last record prior to treatment.

8.3. Analysis Sets

The following analysis sets have been defined for this study:

- Full Analysis Set (FAS): The FAS consists of all participants who initiate the conditioning regimen. This set will include follow-up until Month 15.
- Per Protocol Analysis Set (PPAS): The PPAS consists of all participants in the FAS who complete the IF001 infusion, comply with treatment and follow-up as described in the protocol, and either meet an endpoint or complete the 15-month treatment period with no major protocol deviations
- Extended Analysis Set (EAS): The EAS consists of all participants who have at least one visit between Month >15 and Month 24

The FAS will be used as the efficacy analysis set to evaluate both primary and secondary efficacy endpoints of participants. The FAS will be used for all participant safety analyses, unless otherwise specified. All efficacy and safety endpoints will also be performed using the PPAS. The EAS will be used for secondary extended follow-up analyses.

8.4. Participant Disposition

The number of screen failures, participants enrolled, treated (both initiated therapy and completed the entire regimen), prematurely discontinued from the study, participants in each of the analysis sets, and those with major protocol deviations will be summarized. Major protocol deviations will be summarized and listed by each category. Participants who experience death, graft loss, or remain in the study through Month 15 will be considered to have completed the study.

8.5. Demographic and Other Baseline Characteristics

All demographic and baseline characteristics variables will be listed, and summary statistics provided. Baseline information will be collected during screening. Participant demographics will include age, gender (with childbearing status for females), race, ethnicity, height, weight, and BMI. Other baseline characteristics include relevant medical history (including date of kidney transplantation), current medical conditions, donor characteristics (eg, age, sex, and race), and donor/recipient relationship and match status. Other relevant baseline characteristics will be included in the SAP.

Summary statistics will be presented for the participants in the FAS, PPAS, and EAS. Endpoints with continuous variables will be summarized descriptively (non-missing values, means, standard deviations, medians, 25% and 75% quartiles, minimums, and maximums).

Categorical endpoints will be summarized using the number and percentage within each category (including a category labeled as “missing” when appropriate).

8.6. Concomitant Medications

Concomitant medications will be summarized by therapeutic class and preferred term (Anatomical Therapeutic Chemical levels 2 and 5) by presenting the number and percentage of participants taking each medication.

8.7. Safety Analyses

8.7.1. Safety Variables

Adverse events will be summarized overall for all participants from the time of informed consent onwards. Adverse events will also be summarized by the following study intervals: informed consent to the day prior to IF001 infusion (Day −1), Day 1 to Day 183, and Day 184 to Month 15. The EAS will be summarized for Month >15 to Month 24.

The AE verbatim descriptions (investigator terms from the CRF) will be classified into standardized medical terminology using the Medical Dictionary for Regulatory Activities (MedDRA®) version 28.0 or higher. The incidence of AEs and death will be summarized by system organ class, preferred terms, severity, and relationship to investigational product. The incidence and causes of death will be summarized.

Vital signs and laboratory data will be listed by participant and visit/time. If normal ranges are available, abnormalities will be flagged. Summary statistics will be provided by visit/time.

8.7.2. Primary Safety Analysis

The primary safety endpoint is incidence and severity of AEs, SAEs, and AEs leading to study and/or regimen discontinuation.

Because this is a proof-of-concept study, no hypothesis tests will be conducted on the primary safety endpoint. The primary safety endpoint will be summarized using a point estimate along with 95% CI (exact binomial). The primary analysis will be based on the FAS and will also be performed using the PPAS.

8.7.3. Secondary Safety Analysis

Other incidence rates related to participants' safety profiles will be closely monitored and summarized in DSMB reviews as well as in the final primary analysis. The exposure- (follow-up time) incidence rates and time to event data will be presented up to Day 183, Month 12, and Month 15 as previously described. The list includes:

- The incidence of BK viremia, viruria, infection, and nephropathy
- The incidence of AESI according to defined criteria
- Incidence and grade of acute and severity of chronic GVHD
- Days to neutrophil recovery (>500/µL) and platelet recovery (>20K/µL)
- Number of blood component transfusions
- Incidence of engraftment syndrome
- Incidence of primary and secondary hematopoietic graft failure
- Incidence and grade of bacterial, fungal, and viral infections

8.7.4. Extent of Exposure

Extent of exposure to IF001 will be summarized descriptively. Information that will be summarized include the number of participants exposed as well as total nucleated cells, CD34+ cells and αβ TCR cells in total as well as in million cells per kg of participant IBW. These results will be reported as mean, median, standard deviation, 25% and 75% quartiles, minimum and maximum both as total dose as well as per kg of IBW dose. Participant data listings will be provided for all dosing records and for calculated summary statistics.

8.8. Efficacy Analyses

8.8.1. Primary Efficacy Analysis

The primary efficacy endpoint is the proportion (p) of IF001 participants alive and free from all IS, without graft loss or BPAR, at 15 months post-IF001 infusion. BPAR will be defined based on central pathology with acute cellular ≥1A or antibody-mediated rejection.

Because this is a proof-of-concept study, no hypothesis tests will be conducted on the primary efficacy endpoint. The primary efficacy endpoint will be summarized using a point estimate along with 95% CI (exact binomial).

The primary analysis will be based on the FAS and will also be performed using the PPAS. Participants who are alive and free from all IS without having experienced any BPAR, graft loss, or death at Month 15 will be considered responders. A participant will be deemed to be free from IS at Month 15 if they are not currently taking any IS agents and have not taken IS since complete withdrawal. An exception will be brief use of systemic corticosteroids as described in Section 6.12.

8.8.2. Secondary Efficacy Analysis

The key secondary efficacy variable is the change in renal function (eGFR by CKD-EPI 2021) from the last result prior to the first dose of transplantation conditioning to Month 15.

Because this is a proof-of-concept study, no hypothesis tests will be conducted on the key secondary efficacy endpoint. The mean change in eGFR from baseline to Month 15 will be summarized using descriptive statistics and a 95% CI (exact binomial) will be constructed based on the FAS.

8.8.2.1. Renal Function (eGFR by CKD-EPI 2021)

2021 CKD-EPI Creatinine = 142 × (Scr/A)^{B} × 0.9938$^{\text{age}}$ × (1.012 if female), where A and B are the following:

Female		**Male**	
Scr ≤0.7	$A = 0.7$ $B = -0.241$	**Scr ≤0.9**	$A = 0.9$ $B = -0.302$
Scr >0.7	A = 0.7 B = −1.2	**Scr >0.9**	$A = 0.9$ $B = -1.2$

Abbreviations: Scr, serum creatinine in mg/dL

Imputation for Missing eGFR

For analysis of eGFR, recipients who lost their renal allograft will be assigned a value of zero for their 15-month eGFR. In participants who have not lost their renal allograft, eGFR analysis will be based on Complete Case Analysis.

Additional Analyses

Descriptive summary statistics for eGFR will be repeated and performed for Months 6, 12, and 15.

8.8.2.2. Composite Rate of BPAR, Death, Graft Loss (Renal), or Lost to Follow-up

The proportion of BPAR, death, graft loss, or lost to follow-up will be calculated with 95% CI at Months 6, 12, and 15. The incidence of acute rejection, death, renal allograft loss, or lost to follow-up will be calculated with 95% CI (exact binomial) for those that do not meet the primary efficacy endpoint.

8.8.2.3. Donor Chimerism

The proportion of donor chimerism as defined as ≥50% T-cell chimerism will be calculated with 95% CI (exact binomial) for the FAS. The proportion of participants who were successfully able to wean off IS without BPAR or graft loss at Month 15 among those with donor chimerism will be calculated with 95% CI (exact binomial). Donor chimerism will also be reported as actual percentage for each participant and will be reported for the FAS as mean, median, standard deviation, 25% and 75% quartiles, minimum and maximum, as well as non-missing data (n).

8.8.2.4. Biopsy-Proven Acute Rejection

The proportion of BPAR will be calculated at months 6, 12, and 15. The incidence of BPAR will be summarized by severity, anti-rejection treatment, and steroid-resistance.

8.8.2.5. Donor-Specific Antibodies

The incidence of DSA will be analyzed in the same manner as BPAR.

8.8.2.6. Renal Replacement Therapy

The incidence and duration of renal replacement therapy (hemodialysis) will be analyzed in the same manner as BPAR.

8.8.2.7. Quality of Life Measurements and Variables

Change in quality of life according to SF-36 and ESRD-SCL between the baseline assessment and Months 12 and 15 will be described. Details regarding the approach will be provided in the SAP.

8.9. Sample Size

The planned sample size of 15 participants was empirically determined and will provide adequate safety, tolerability, and preliminary efficacy data to achieve study objectives while exposing as few participants as possible to the study procedures.

8.10. Handling of Missing Data

Efforts will be taken to limit premature discontinuations and ensure completeness of data collection. Analyses will utilize participant data which is collected; no imputation of missing data will be performed except as described in Section 8.8.2.1. All participants with data for a specific analysis will be included; there are no analyses that will exclude participants regardless of the amount or type of data missing.

8.11. Subgroup Analyses

No subgroup analyses are planned in this study.

8.12. Interim Analyses

No interim analyses are planned.

9. Data Safety Monitoring Board

An independent DSMB will be instituted at study initiation and will advise the sponsor and investigators on matters relating to the safety and conduct of the study. The DSMB will be governed by a charter that will include meeting frequency, stopping rules, and event triggers necessitating an unscheduled review. Further enrollment into the study will be temporarily halted by the sponsor for a full DSMB safety assessment if:

- Two out of the first 10 participants or ≥20% of participants subsequently at any time during the trial develop severe (ie, Grade 3-4) acute GVHD, or
- More than 2 out of the first 10 participants or >20% of participants at any time during the trial subsequently experience BPAR (local pathology), or
- Three out of the first 10 participants or ≥30% of participants subsequently at any time during the study develop:
    - Grade 3-4 infusion reactions to IF001, or
    - Grade 4 neutropenia lasting more than 28 days, or
    - Grade 4 organ toxicity within the first 30 days, or
- Any death or graft loss occurs

The DSMB will critically assess the background and clinical course of events leading to the temporary halt of the study as soon as possible.

10. Study Administration

10.1. Inspection of Records

The sponsor will be allowed to conduct site visits to the investigation facilities for the purpose of monitoring any aspect of the study. The investigator agrees to allow the monitor to inspect the investigational product storage area and accountability records, participant charts, study source documents, and other records relative to study conduct.

10.2. Site Monitoring

Before the study starts, at a site initiation visit or at an investigator's meeting, a sponsor representative and/or designee will review the protocol and CRFs with the investigators and their staff. During the study, the sponsor will ensure compliance with the protocol and Good Clinical Practice (GCP) as well as the quality and integrity of the data at each site. The Clinical Research Associate (CRA) will visit the site to check the completeness of participant records, the accuracy of entries on the CRFs, the adherence to the protocol, and to GCP, the progress of enrollment, and to ensure that investigational product is being stored, administered, and accounted for according to specifications.

Key study personnel must be available to assist the CRA during these visits. Continuous remote monitoring of each site's data may be performed in addition to the on-site visits, if allowed by the site. Additionally, ongoing safety reviews may be performed to identify risks and trends for site operational parameters, and to assist with trial oversight.

The investigator must maintain source documents for each participant in the study. These include case and visit notes (hospital or clinic medical records) containing demographic and medical information, laboratory data, ECGs, and the results of any other tests or assessments. All information on CRFs must be traceable to these source documents in the participant files. The investigator must also keep all original informed consent documents signed by the participant (a signed copy is given to the participant; see Section 13.2).

The CRA must have access to all relevant source documents to confirm their consistency with the CRF entries. The study monitoring standards require full verification for the presence of informed consent, adherence to study eligibility criteria, and documentation of SAEs.

Monitoring of the source data and CRFs will be performed according to the study-specific monitoring plan. The monitor will be available between visits if the investigator(s) or other staff need information or advice.

10.2.1. **Study Completion and Site Closure**

Upon completion of the study, the monitor will conduct the following activities in conjunction with the investigator or study site personnel, as appropriate:

- Return of all study data to the sponsor
- Data queries
- Accountability and reconciliation of investigational product
- Review of study records for completeness
- Reconciliation of Investigator Site File and Trial Master File

10.3. **Access to Information for Audits and Inspections**

Authorized representatives of the sponsor, a regulatory authority, or an IRB may visit the site to perform audits or inspections, including source data verification. The purpose of a sponsor audit or inspection is to systematically and independently examine all study-related activities and documents to determine whether these activities were conducted, and data were recorded, analyzed, and accurately reported according to the protocol, GCP, and any applicable regulatory requirements. The investigator should contact the sponsor immediately if contacted by a regulatory agency about an inspection.

10.4. **Retention of Records**

The Principal Investigator must maintain all documentation relating to the study for a period of 2 years after the last marketing application approval, or if not approved, 2 years following the discontinuance of the test article for investigation. If it becomes necessary for the sponsor or a regulatory authority to review any documentation relating to the study, the investigator must permit access to such records.

10.5. Publication and Data Sharing Policy

No data collected as part of the study will be utilized in any written work, including publications, without the written consent of the sponsor. The sponsor will post key design elements of this protocol in a publicly accessible database such as clinicaltrials.gov. Upon study completion and finalization of the study report, the results of the trial will be either submitted for publication and/or posted in a publicly accessible database of clinical trial results.

11. Data Collection

11.1. Data System

The data system being used for this study is Advantage eClinical® managed by Emmes, contracted by the sponsor. The Principal Investigator and at least 2 staff from each site will need to complete training requirements and obtain access to this electronic data capture system.

11.2. Case Report Forms

11.2.1. Criteria for Forms Submission

Criteria for timeliness of submission for all study forms are detailed in the eCRF Completion Guide and Advantage eClinical User's Guide. Forms that are not entered into Advantage eClinical within the specified time will be considered delinquent. A missing form will continue to be requested either until the form is entered into Advantage eClinical and integrated into the master database at Emmes, or until an exception is granted by Emmes staff and entered into the Missing Form Exception File, as detailed in the Advantage eClinical User's Guide.

11.2.2. Reporting Participant Deaths

Participant death information must be entered into Advantage eClinical within 24 business hours of knowledge of the participant's death. If the cause of death is unknown at that time, it does not need to be recorded at that time. However, once the cause of death is determined, the Death Form must be updated in Advantage eClinical. See additional SAE reporting information in Section 7.4.2.

11.2.3. Center for International Blood and Marrow Transplant Research Data Reporting

Centers participating in this study must register pre- and post-HCT outcomes to the CIBMTR. Registration is done using procedures and forms of the Stem Cell Transplant Outcomes Database. (Note: Federal legislation requires submission of these forms for all United States allotransplant recipients). The CIBMTR forms will be submitted directly to the CIBMTR through routine CIBMTR mechanisms.

12. Quality Assurance and Quality Control

12.1. Data Quality Assurance

Sponsor staff and/or its designee review the data entered in the CRFs by investigational site staff for completeness and accuracy and instruct the site personnel to make any required corrections or additions. Site staff are required to respond to any queries received and confirm or correct the data.

Concomitant medications entered into the database will be coded using the World Health Organization Drug Reference List, which employs the Anatomical Therapeutic Chemical classification system.

Concomitant procedures, non-drug therapies, and AEs will be coded using MedDRA (version 28.0 or higher) terminology.

Results from laboratory samples tested centrally will be sent electronically to the sponsor and/or its designee.

The occurrence of protocol deviations will be reported. After these actions have been completed and the database has been declared to be complete and accurate, it will be locked and made available for data analysis. Any changes to the database after that time can only be made after written agreement by the sponsor clinical team.

12.2. Protocol Adherence

This protocol defines the study objectives, the study procedures, and the data to be collected on study participants. Additional assessments required to ensure safety of participants should be conducted if considered necessary on a case-by-case basis. Under no circumstances should an investigator collect additional data or conduct any additional procedures for any research-related purposes that are not part of standard of care involving any investigational treatments.

Investigators ascertain they will apply due diligence to avoid protocol deviations; all significant deviations will be recorded and reported in the clinical study report. If an investigator believes that a protocol deviation would, for only safety reasons, improve the conduct of the study and would be in the best interest of the participant, this must be considered a protocol amendment. Unless such an amendment is agreed upon by the sponsor and the necessary approvals obtained, it cannot be implemented.

12.3. Protocol Amendments

Any change or addition to the protocol can only be made in a written protocol amendment that must be approved by the sponsor and the IRB prior to implementation. Only amendments that are intended to eliminate an apparent immediate hazard to participants can be implemented immediately, provided health authorities are subsequently notified by protocol amendment and the reviewing IRB is notified.

Notwithstanding the need for approval of formal protocol amendments, the investigator is expected to take any immediate action required for the safety of any participant in this study, even if the action represents a deviation from the protocol. In such cases, the reporting requirements identified in Section 7 should be followed.

13. Ethics

This clinical study was designed and will be implemented and reported in accordance with ethical principles that have their origin in the Declaration of Helsinki, International Conference on Harmonization GCP, and all applicable regulatory requirements.

13.1. Institutional Review Board

Before the study starts, the investigator and institution must obtain IRB approval for the protocol, ICF, recruitment materials and procedures (such as advertisements), and any other written information provided to potential participants. Written IRB approval must be submitted to the sponsor prior to enrolling a participant into the study. Initial IRB approval, and all materials approved by the IRB for this study including the participant ICF and recruitment materials must be maintained by the investigator and made available for inspection.

Prior to study start, the Principal Investigator must sign the Investigator's Agreement (Appendix 7). The Principal Investigator is also responsible for informing the IRB any amendment to the protocol in accordance with local requirements. The protocol must be re-approved by the IRB upon receipt of amendments and annually, as local regulations require.

13.2. Informed Consent

The Principal Investigator will ensure that the patient is given full and adequate oral and written information about the nature, purpose, and possible risks and benefits of the study. Patients must also be notified that they are free to discontinue from the study at any time. The patient should be given the opportunity to ask questions and allowed time to consider the information provided.

Women of childbearing potential should be informed that the study treatment may involve unknown risks to the fetus if pregnancy were to occur during the study and agree that to participate in the study, they must adhere to the contraception requirement as described in Section 4.2. If there is any question that the patient will not reliably comply, they should not be enrolled in the study.

The participant's signed and dated informed consent must be obtained before conducting any study procedures.

The Principal Investigator must maintain the original, signed ICF and give a copy of the signed ICF to the participant. The process of obtaining informed consent should be documented in the participant source documents.

Donor informed consent will be obtained by the NMDP.

13.3. Participant and Data Confidentiality

After the ICF has been signed, each participant will be assigned a unique identifier as described in Section 5.2. To ensure confidentiality of records and personal data of participants, any participant records or datasets transferred to the sponsor will only include the identifier. The names of participants or any information that would make the participant identifiable will not be transferred.

The participant must be informed that his/her personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the participant.

The participant must be informed that his/her medical records (eg, CRFs) recorded during the study may be examined by research quality assurance auditors or other authorized personnel appointed by the sponsor, by appropriate IRB members, and by inspectors from regulatory authorities.

Placeholder for company-specific wording regarding data protection policies regarding data breaches etc.

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15. APPENDICES

APPENDIX 1 REVACCINATION SCHEDULE

Immunizations After HCT or High-Dose Cyclophosphamide

Time after HCT or high-dose Cy (months)	**COVID-19[1]**	**Influenza (inactivated)[2]**	**PCV13**	**PPSV23[3]**	**Hib**	**DTaP[4]**	**HBV[5]**	**IPV**	**HPV**	**RZV**	**MMR[6]**
3	X										
4	X										
6		X	X		X	X		X	X		
8			X		X	X	X	X	X		
10			X			X				X	
12				X	X		X	X	X	X	
24											X

Abbreviations: COVID-19, Coronavirus disease 2019; Cy, cyclophosphamide; DTaP, diphtheria, tetanus toxoids, and acellular pertussis vaccine; HBV, hepatitis B vaccine; Hib, Haemophilus b conjugate vaccine; HCT, hematopoietic cell transplantation; HPV, human papillomavirus vaccine; IPV, inactivated polio vaccine; IS, immunosuppression; MMR, measles, mumps, and rubella vaccine; Td, diptheria and tetanus toxoid; Tdap, diphtheria, tetanus toxoids, and acellular pertussis vaccine; PCV13, pneumococcal conjugate vaccine (Prevnar®); PPSV23, pneumococcal polysaccharide vaccine (Pneumovax®); RZV, zoster vaccine recombinant vaccine (Shingrix)

[1] Pfizer-BioNTech (SpikeVax) is a 2-dose series given 3 weeks apart; Moderna (Comirnaty) is a 2-dose series given 4 weeks apart; Janssen is given as 1 dose. Follow the current recommendations from the Centers for Disease Control and Prevention regarding booster doses and timing.

[2] May be given no sooner than 4 months after HCT or high-dose cyclophosphamide in the event of a community outbreak. If influenza vaccine was given <6 months after HCT or high-dose cyclophosphamide or if the participant was on IS when first vaccinated, consider an additional dose after the participant is off IS for 2 months.

[3] PPSV23 should not be given to participants with chronic GVHD; a fourth dose of PCV13 may be given instead. PPSV23 must be given 2 months after PCV13.

[4] Alternatively, a dose of Tdap should be administered followed by either 2 doses of DT or 2 doses of Td.

[5] For participants aged 18 to 45 years.

[6] Do not administer to participants with active GHVD or ongoing IS. Live vaccines should be given 8 to 11 months after the last dose of intravenous immunoglobulin.

Source: Rubin 2014; Khawaja 2022

Appendix 2 Acute GVHD Staging (MAGIC Criteria)

GVHD Target Organ Staging

Stage	Skin (Active Erythema Only)	Liver (Bilirubin)	Upper GI	Lower GI (stool output/day)
0	No active (erythematous) GVHD rash	<2 mg/dL	No or intermittent nausea, vomiting, or anorexia	Adult: <500 mL/day or <3 episodes/day Child: <10 mL/kg/day or <4 episodes/day
1	Maculopapular rash <25% BSA	2-3 mg/dL	Persistent nausea, vomiting or anorexia	Adult: 500-999 mL/day or 3-4 episodes/day Child: 10-19.9 mL/kg/day or 4-6 episodes/day
2	Maculopapular rash 25-50% BSA	3.1-6 mg/dL		Adult: 1000-1500 mL/day or 5-7 episodes/day Child: 20-30 mL/kg/day or 7-10 episodes/day
3	Maculopapular rash >50% BSA	6.1-15 mg/dL		Adult: >1500 mL/day or >7 episodes/day Child: >30 mL/kg/day or >10 episodes/day
4	Generalized erythroderma (>50% BSA) *plus* bullous formation and desquamation >5% BSA	>15 mg/dL		Severe abdominal pain with or without ileus or grossly bloody stool (regardless of stool volume).

Overall clinical grade (based on most severe target organ involvement):
Grade 0: No stage 1-4 of any organ.
Grade I: Stage 1-2 skin without liver, upper GI, or lower GI involvement.
Grade II: Stage 3 rash and/or stage 1 liver and/or stage 1 upper GI and/or stage 1 lower GI.
Grade III: Stage 2-3 liver and/or stage 2-3 lower GI, with stage 0-3 skin and/or stage 0-1 upper GI.
Grade IV: Stage 4 skin, liver, or lower GI involvement, with stage 0-1 upper GI.

Source: Table 1, Harris 2016

Appendix 3 NIH Chronic GVHD Diagnosis and Staging Consensus Recommendations

	SCORE 0	SCORE 1	SCORE 2	SCORE 3
Performance Score: KPS ECOG LPS	☐ Asymptomatic and fully active (ECOG 0; KPS or LPS 100%)	☐ Symptomatic, fully ambulatory, restricted only in physically strenuous activity (ECOG 1, KPS or LPS 80-90%)	☐ Symptomatic, ambulatory, capable of self-care, >50% of waking hours out of bed (ECOG 2, KPS or LPS 60-70%)	☐ Symptomatic, limited self-care, >50% of waking hours in bed (ECOG 3–4, KPS or LPS <60%)
SKIN† **Score % BSA** *GVHD features to be scored by BSA:* **Check all that apply:** ☐ Maculopapular rash/erythema ☐ Lichen planus-like features ☐ Sclerotic features ☐ Papulosquamous lesions or ichthyosis ☐ Keratosis pilaris-like GVHD	☐ No BSA involved	☐ 1-18% BSA	☐ 19-50% BSA	☐ >50% BSA
Skin features score:	☐ No sclerotic features		☐ Superficial sclerotic features "not hidebound" (able to pinch)	**Check all that apply:** ☐ Deep sclerotic features ☐ "Hidebound" (unable to pinch) ☐ Impaired mobility ☐ Ulceration
Other skin GVHD features (NOT scored by BSA) **Check all that apply:** ☐ Hyperpigmentation ☐ Hypopigmentation ☐ Poikiloderma ☐ Severe or generalized pruritus ☐ Hair involvement ☐ Nail involvement ☐ *Abnormality present but explained entirely by non-GVHD documented cause (specify):*___				
Mouth *Lichen planus-like features present:* ☐ **Yes** ☐ **No**	☐ No symptoms	☐ Mild symptoms **with** disease signs but not limiting oral intake significantly	☐ Moderate symptoms with disease signs **with** partial limitation of oral intake	☐ Severe symptoms with disease signs on examination **with** major limitation of oral intake
☐ *Abnormality present but explained entirely by non-GVHD documented cause (specify):*___				

Figure 1. Organ scoring of chronic GVHD. ECOG indicates Eastern Cooperative Oncology Group; KPS, Karnofsky Performance Status; LPS, Lansky Performance Status; BSA, body surface area; ADL, activities of daily living; LFTs, liver function tests; AP, alkaline phosphatase; ALT, alanine aminotransferase; ULN, normal upper limit. *Weight loss within 3 months. †Skin scoring should use both percentage of BSA involved by disease signs and the cutaneous features scales. When a discrepancy exists between the percentage of total body surface (BSA) score and the skin feature score, OR if superficial sclerotic features are present (Score 2), but there is impaired mobility or ulceration (Score 3), the higher level should be used for the final skin scoring. ‡To be completed by specialist or trained medical providers (see Supplemental Figure). **Lung scoring should be performed using both the symptoms and FEV1 scores whenever possible. FEV1 should be used in the final lung scoring where there is discrepancy between symptoms and FEV1 scores.

	SCORE 0	SCORE 1	SCORE 2	SCORE 3
EYES *Keratoconjunctivitis sicca (KCS) confirmed by ophthalmologist:* ☐ **Yes** ☐ **No** ☐ **Not examined**	☐ No symptoms	☐ Mild dry eye symptoms not affecting ADL (requirement of lubricant eye drops ≤ 3 x per day)	☐ Moderate dry eye symptoms partially affecting ADL (requiring lubricant eye drops > 3 x per day or punctal plugs), **WITHOUT** new vision impairment due to KCS	☐ Severe dry eye symptoms significantly affecting ADL (special eyeware to relieve pain) **OR** unable to work because of ocular symptoms **OR** loss of vision due to KCS
☐ *Abnormality present but explained entirely by non-GVHD documented cause (specify):*				
GI Tract ***Check all that apply:*** ☐ Esophageal web/ proximal stricture or ring ☐ Dysphagia ☐ Anorexia ☐ Nausea ☐ Vomiting ☐ Diarrhea ☐ Weight loss ≥5%* ☐ Failure to thrive	☐ No symptoms	☐ Symptoms without significant weight loss* (<5%)	☐ Symptoms associated with mild to moderate weight loss* (5-15%) **OR** moderate diarrhea without significant interference with daily living	☐ Symptoms associated with significant weight loss* >15%, requires nutritional supplement for most calorie needs **OR** esophageal dilation **OR** severe diarrhea with significant interference with daily living
☐ *Abnormality present but explained entirely by non-GVHD documented cause (specify):*____________________				
LIVER	☐ Normal total bilirubin and ALT or AP < 3 x ULN	☐ Normal total bilirubin with ALT ≥3 to 5 x ULN or AP ≥ 3 x ULN	☐ Elevated total bilirubin but ≤3 mg/dL or ALT > 5 ULN	☐ Elevated total bilirubin > 3 mg/dL
☐ *Abnormality present but explained entirely by non-GVHD documented cause (specify):*				
LUNGS** **Symptom score**:	☐ No symptoms	☐ Mild symptoms (shortness of breath after climbing one flight of steps)	☐ Moderate symptoms (shortness of breath after walking on flat ground)	☐ Severe symptoms (shortness of breath at rest; requiring 0_2)
Lung score: **% FEV1** ☐ *Pulmonary function tests* ☐ Not performed	☐ FEV1≥80%	☐ FEV1 60-79%	☐ FEV1 40-59%	☐ FEV1 ≤39%
☐ *Abnormality present but explained entirely by non-GVHD documented cause (specify):*____________________				

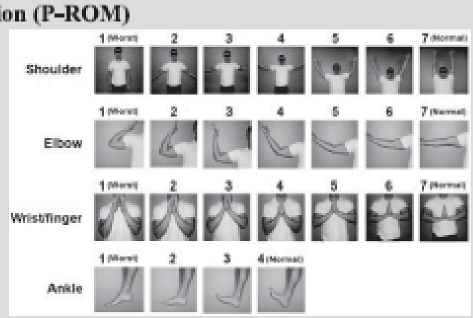
	SCORE 0	SCORE 1	SCORE 2	SCORE 3
JOINTS AND FASCIA P-ROM score *(see below)* Shoulder (1-7):___ Elbow (1-7): ____ Wrist/finger (1-7):___ Ankle (1-4):___	☐ No symptoms	☐ Mild tightness of arms or legs, normal or mild decreased range of motion (ROM) **AND** not affecting ADL	☐ Tightness of arms or legs **OR** joint contractures, erythema thought due to fasciitis, moderate decrease ROM **AND** mild to moderate limitation of ADL	☐ Contractures **WITH** significant decrease of ROM ***AND*** significant limitation of ADL (unable to tie shoes, button shirts, dress self etc.)
☐ *Abnormality present but explained entirely by non-GVHD documented cause (specify):*___				
GENITAL TRACT (*See Supplemental figure*‡) ☐ Not examined *Currently sexually active* ☐ Yes ☐ No	☐ No signs	☐ Mild signs‡ and females with or without discomfort on exam	☐ Moderate signs‡ and may have symptoms with discomfort on exam	☐ Severe signs‡ with or without symptoms
☐ *Abnormality present but explained entirely by non-GVHD documented cause (specify):*___				

Other indicators, clinical features or complications related to chronic GVHD (check all that apply and assign a score to severity (0-3) based on functional impact where applicable none – 0,mild -1, moderate -2, severe – 3)

☐ Ascites (serositis)___	☐ Myasthenia Gravis___	
☐ Pericardial Effusion___	☐ Peripheral Neuropathy___	☐ Eosinophilia > 500/μl___
☐ Pleural Effusion(s)___	☐ Polymyositis___	☐ Platelets <100,000/μl ___
☐ Nephrotic syndrome___	☐ Weight loss>5%* without GI symptoms ___	☐ Others (specify):___

Overall GVHD Severity *(Opinion of the evaluator)*	☐ **No GVHD**	☐ **Mild**	☐ **Moderate**	☐ **Severe**

Photographic Range of Motion (P-ROM)



Source: Figure 1, Jagasia 2015

Appendix 4 Banff 2022 Criteria

CATEGORY 1: Normal Biopsy Or Nonspecific Changes
Requires exclusion of any diagnosis from the Banff Diagnostic Categories 2-4, 6 below.

CATEGORY 2: Antibody-mediated rejection (AMR) and microvascular inflammation/injury (AMR/MVI)

Active AMR; All 3 criteria must be met for diagnosis

1. Active lesions* of AMR present, at least 1 of the following:
   - Microvascular inflammation (g > 0 and/or peritubular capillaries [ptc] > 0), in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute T Cell-Mediated Rejection (TCMR), borderline infiltrate, or infection, ptc ≥ 1 alone is not sufficient and g must be ≥ 1
   - Intimal or transmural arteritis (v > 0)
   - Acute thrombotic microangiopathy, in the absence of any other cause
2. At least 1 or more of the following:
   - Linear C4d staining in peritubular capillaries or medullary vasa recta (C4d2 or C4d3 by immunofluorescence (IF) on frozen sections, or C4d > 0 by immunohistochemistry (IHC) on paraffin sections)
   - At least moderate microvascular inflammation ([g + ptc] ≥2) in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, ptc ≥ 2 alone is not sufficient and g must be ≥1
   - Biopsy-based transcript diagnostics for AMR/MVI above a defined threshold, if thoroughly validated for use as substitute for MVI and available
3. Evidence of circulating donor-specific antibodies (DSA to human leukocyte antigen [HLA] or other antigens). If thorough testing for DSA (anti-HLA or other specificity) has not yet been performed, this should be done, following the STAR guidelines. Detection of non-HLA antibodies (including ABO antibodies in ABO-incompatible transplantation) can be used as serologic Banff criterion for diagnosis of AMR, if the testing protocols are sufficiently standardized and clinically validated for the appropriate clinical context. C4d staining as noted above in Criterion 2 may substitute for DSA.

*Can be observed in AMR and strengthen the diagnosis but not diagnostic in itself: acute tubular injury, in the absence of any other apparent cause

Chronic active AMR; all 3 criteria must be met for diagnosis

1. Chronic lesions* of AMR present, at least 1 of the following:
   - Transplant glomerulopathy (cg > 0) if no evidence of chronic thrombotic microangiopathy (TMA) or chronic recurrent/de novo glomerulonephritis; includes changes evident by electron microscopy (EM) alone (cg1a)
   - Severe peritubular capillary basement membrane multilayering (requires EM)
2. Identical to criterion 2 for active AMR, above
3. Identical to criterion 3 for active AMR, above, including strong recommendation for DSA testing whenever criteria 1 and 2 are met.

*Other lesions can be observed in AMR and strengthen the diagnosis but are not diagnostic by themselves: arterial intimal fibrosis (cv) of new onset, excluding other causes; leukocytes within the sclerotic intima favour chronic AMR if there is no prior history of TCMR;
Chronic AMR; all 3 criteria must be met for diagnosis
1. cg > 0 and/or severe ptcml 2. Absence of criterion 2 as defined for active and chronic active AMR, above 3. Prior documented diagnosis of active or chronic active antibody-mediated rejection (ABMR) and/or documented prior (post-transplant) and/or current evidence of DSA (DSA as defined in above criterion 3 for active AMR)
C4d staining without evidence of rejection; all 4 features must be met for diagnosis[c]
1. Linear C4d staining in peritubular capillaries (C4d2 or C4d3 by IF on frozen sections, or C4d > 0 by IHC on paraffin sections) 2. Criterion 1 for active or chronic active AMR not met 3. Negative biopsy-based transcript diagnostics for AMR/MVI as in criterion 2 for active and chronic active AMR 4. No acute or chronic active TCMR, or borderline changes
Microvascular inflammation/injury (MVI), DSA-negative and C4d-negative; all 3 criteria must be met for diagnosis
1. At least moderate microvascular inflammation ([g + ptc] ≥2) in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, ptc ≥ 2 alone is not sufficient and g must be ≥1 2. No linear C4d staining in peritubular capillaries (C4d0 or C4d1 by IF on frozen sections, or C4d = 0 by IHC on paraffin sections) 3. No serologic evidence of circulating donor-specific antibodies (DSA to HLA or other antigens, as defined in above criterion 3 for active AMR)
Probable AMR; all 4 criteria must be met for diagnosis
1. Identical to criterion 1 for active AMR, above 2. Criterion 1 for chronic active and chronic AMR not met 3. Absence of criterion 2 defined for active and chronic active AMR, above **4.** Identical to criterion 3 for active AMR, above (but C4d must be negative)
C4d staining with acute tubular injury (ATI); all 3 criteria must be met for diagnosis
1. Acute tubular injury (ATI) is present 2. Linear C4d staining in peritubular capillaries (C4d2 or C4d3 by IF on frozen sections, or C4d > 0 by IHC on paraffin sections) 3. Criterion 1 for active or chronic active AMR not met Clinical scenarios: ● Early posttransplant in crossmatch positive DSA sensitised patient -> “Probable AMR” ● ABO-incompatibility -> “Accommodation” ● DSA negative in conventional transplants -> “No AMR”

CATEGORY 3: Suspicious (Borderline) For Acute TCMR Foci of Banff Lesion Score t>0 AND Banff Lesions Score i=1 OR Foci of Banff Lesion Score t1 AND Banff Lesion Score i ≥ 2
CATEGORY 4: TCMR
Acute TCMR IA Banff Lesion Score i ≥ 2 AND Banff Lesion Score t2
Acute TCMR IB Banff Lesion Score i ≥ 2 AND Banff Lesion Score t3
Acute TCMR IIA Banff Lesion Score v1 regardless of Banff Lesion Scores i or t
Acute TCMR IIB Banff Lesion Score v2 regardless of Banff Lesion Scores i or t
Acute TCMR III Banff Lesion Score v3 regardless of Banff Lesion Scores i or t
Chronic Active TCMR Grade IA Banff Lesion Score ti ≥ 2 AND Banff Lesion Score i-IFTA ≥ 2, other known causes of i-IFTA (eg, pyelonephritis, BK-virus nephritis etc.) ruled out AND Banff Lesion Score t2
Chronic Active TCMR Grade IB Banff Lesion Score ti ≥ 2 AND Banff Lesion Score i-IFTA ≥ 2, other known causes of i-IFTA ruled out AND Banff Lesion Score t3
Chronic Active TCMR Grade II Arterial intimal fibrosis with mononuclear cell inflammation in fibrosis and formation of neointima
CATEGORY 5: IFTA
Grade I (Mild) Banff Lesion Score ci1 OR Banff Lesion Score ct1 **Grade II** (Moderate) Banff Lesion Score ci2 OR Banff Lesion Score ct2 **Grade III** (Severe) Banff Lesion Score ci3 OR Banff Lesion Score ct3
CATEGORY 6: Other Changes Not Considered To Be Caused By Acute Or Chronic Rejection
Polyomavirus Nephropathy, Posttransplant Lymphoproliferative Disorder, Calcineurin Inhibitor Toxicity, Acute Tubular Injury, Recurrent Disease, De Novo Glomerulopathy (Other Than TG), Pyelonephritis, Drug-Induced Interstitial Nephritis

References: Naesens 2024; Loupy 2020; Haas 2018

Appendix 5 SF-36 Rating scale

The 36-Item Health Survey was developed at RAND as part of the Medical Outcomes Study (RAND 2025).

RAND 36-Item Health Survey 1.0 Questionnaire Items

Choose one option for each questionnaire item.

1. In general, would you say your health is:

○ 1 - Excellent

○ 2 - Very good

○ 3 - Good

○ 4 - Fair

○ 5 - Poor

---

2. **Compared to one year ago**, how would you rate your health in general **now**?

○ 1 - Much better now than one year ago

○ 2 - Somewhat better now than one year ago

○ 3 - About the same

○ 4 - Somewhat worse now than one year ago

○ 5 - Much worse now than one year ago

The following items are about activities you might do during a typical day. Does **your health now limit you** in these activities? If so, how much?

	Yes, limited a lot	Yes, limited a little	No, not limited at all
3. **Vigorous activities**, such as running, lifting heavy objects, participating in strenuous sports	○ 1	○ 2	○ 3
4. **Moderate activities**, such as moving a table, pushing a vacuum cleaner, bowling, or playing golf	○ 1	○ 2	○ 3
5. Lifting or carrying groceries	○ 1	○ 2	○ 3
6. Climbing **several** flights of stairs	○ 1	○ 2	○ 3
7. Climbing **one** flight of stairs	○ 1	○ 2	○ 3
8. Bending, kneeling, or stooping	○ 1	○ 2	○ 3
9. Walking **more than a mile**	○ 1	○ 2	○ 3
10. Walking **several blocks**	○ 1	○ 2	○ 3
11. Walking **one block**	○ 1	○ 2	○ 3
12. Bathing or dressing yourself	○ 1	○ 2	○ 3

During the **past 4 weeks**, have you had any of the following problems with your work or other regular daily activities **as a result of your physical health**?

	Yes	No
13. Cut down the **amount of time** you spent on work or other activities	○ 1	○ 2
14. **Accomplished less** than you would like	○ 1	○ 2
15. Were limited in the **kind** of work or other activities	○ 1	○ 2
16. Had **difficulty** performing the work or other activities (for example, it took extra effort)	○ 1	○ 2

During the **past 4 weeks**, have you had any of the following problems with your work or other regular daily activities **as a result of any emotional problems** (such as feeling depressed or anxious)?

	Yes	No
17. Cut down the **amount of time** you spent on work or other activities	○ 1	○ 2
18. **Accomplished less** than you would like	○ 1	○ 2
19. Didn't do work or other activities as **carefully** as usual	○ 1	○ 2

---

20. During the **past 4 weeks**, to what extent has your physical health or emotional problems interfered with your normal social activities with family, friends, neighbors, or groups?

○ 1 - Not at all

○ 2 - Slightly

○ 3 - Moderately

○ 4 - Quite a bit

○ 5 - Extremely

---

21. How much **bodily** pain have you had during the **past 4 weeks**?

○ 1 - None

○ 2 - Very mild

○ 3 - Mild

○ 4 - Moderate

○ 5 - Severe

○ 6 - Very severe

---

22. During the **past 4 weeks**, how much did **pain** interfere with your normal work (including both work outside the home and housework)?

- ○ 1 - Not at all
- ○ 2 - A little bit
- ○ 3 - Moderately
- ○ 4 - Quite a bit
- ○ 5 - Extremely

These questions are about how you feel and how things have been with you **during the past 4 weeks**. For each question, please give the one answer that comes closest to the way you have been feeling.

How much of the time during the **past 4 weeks**...

	All of the time	Most of the time	A good bit of the time	Some of the time	A little of the time	None of the time
23. Did you feel full of pep?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6
24. Have you been a very nervous person?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6
25. Have you felt so down in the dumps that nothing could cheer you up?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6
26. Have you felt calm and peaceful?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6
27. Did you have a lot of energy?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6
28. Have you felt downhearted and blue?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6
29. Did you feel worn out?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6
30. Have you been a happy person?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6
31. Did you feel tired?	○ 1	○ 2	○ 3	○ 4	○ 5	○ 6

32. During the **past 4 weeks**, how much of the time has **your physical health or emotional problems** interfered with your social activities (like visiting with friends, relatives, etc.)?

- ○ 1 - All of the time
- ○ 2 - Most of the time
- ○ 3 - Some of the time
- ○ 4 - A little of the time
- ○ 5 - None of the time

How TRUE or FALSE is **each** of the following statements for you.

	Definitely true	Mostly true	Don't know	Mostly false	Definitely false
33. I seem to get sick a little easier than other people	○ 1	○ 2	○ 3	○ 4	○ 5
34. I am as healthy as anybody I know	○ 1	○ 2	○ 3	○ 4	○ 5
35. I expect my health to get worse	○ 1	○ 2	○ 3	○ 4	○ 5
36. My health is excellent	○ 1	○ 2	○ 3	○ 4	○ 5

Appendix 6 ESRD-SCL Rating Scale

ESRD-SCL-TM

End-Stage Renal Disease – Symptom Checklist – Transplantation Module

Bereitgestellt von PSYCHOMETRIKON Psychologisch-medizinisches Testportal

Code ______ Date (D D M M Y Y) ______ Time __ : __

Below there is a list of physical and/or mental problems you might experience. Please tick the number that best describes your current condition. Please answer each of the questions!

Please enter: Date of Birth (D D M M Y Y) ______ Sex m f

	How much are you currently suffering from …	**Not at all**	**Somewhat**	**Moderately**	**Strongly**	**Extremely**
1	poor general health	0	1	2	3	4
2	limited physical performance	0	1	2	3	4
3	your appearance	0	1	2	3	4
4	limited mental abilities	0	1	2	3	4
5	thoughts about a/my transplantation	0	1	2	3	4
6	brooding about the organ donor	0	1	2	3	4
7	asking yourself how long the transplant will be working effectively	0	1	2	3	4
8	nightmares	0	1	2	3	4
9	headaches	0	1	2	3	4
10	sleeplessness	0	1	2	3	4
11	irritability	0	1	2	3	4
12	concentration difficulties	0	1	2	3	4
13	nervousness	0	1	2	3	4
14	severe dizziness	0	1	2	3	4
15	anxiety	0	1	2	3	4

ESRD-SCL-TM

End-Stage Renal Disease – Symptom Checklist – Transplantation Module

Bereitgestellt von PSYCHOMETRIKON Psychologisch-medizinisches Testportal

Code	Date (D D M M Y Y)	Time	:

	How much are you currently suffering from ...	Not at all	Somewhat	Moderately	Strongly	Extremely
16	forgetfulness	0	1	2	3	4
17	impaired visual ability	0	1	2	3	4
18	impaired hearing	0	1	2	3	4
19	ear ringing or buzzing	0	1	2	3	4
20	mood swings	0	1	2	3	4
21	irregular heartbeat	0	1	2	3	4
22	increased blood pressure	0	1	2	3	4
23	bone pain	0	1	2	3	4
24	joint pain	0	1	2	3	4
25	muscle pain	0	1	2	3	4
26	colds or influenza	0	1	2	3	4
27	increased body hair	0	1	2	3	4
28	gum proliferation	0	1	2	3	4
29	gum bleeding	0	1	2	3	4
30	a puffy face	0	1	2	3	4

ESRD-SCL-TM

End-Stage Renal Disease – Symptom Checklist – Transplantation Module

Bereitgestellt von PSYCHOMETRIKON Psychologisch-medizinisches Testportal

Code [ ] [ ] [ ] [ ] [ ] [ ] Date D D M M Y Y Time [ ] [ ] : [ ] [ ]

	How much are you currently suffering from ...	Not at all	Somewhat	Moderately	Strongly	Extremely
31	proneness to infections	0	1	2	3	4
32	swollen feet	0	1	2	3	4
33	stomach ache	0	1	2	3	4
34	disturbance of sensitivity in your legs	0	1	2	3	4
35	a feeling of being exhausted	0	1	2	3	4
36	gum changes	0	1	2	3	4
37	increased hair growth	0	1	2	3	4
38	a puffy face in the morning	0	1	2	3	4
39	weight change	0	1	2	3	4
40	swollen legs	0	1	2	3	4
41	increased bruising	0	1	2	3	4
42	increased thirst	0	1	2	3	4
43	memory disturbances	0	1	2	3	4

Appendix 7 Investigator's Agreement

I have read IF001-CL201 protocol version 1.0 and agree to conduct the study as outlined, and in accordance with the guidelines and all applicable government regulations. I agree to maintain the confidentiality of all information received or developed in connection with this protocol.

____

Printed Name of Investigator

____

Signature of Investigator

____

Date

Appendix 8 Sponsor Signature Page

Sponsor's Approval

The protocol has been approved by ImmunoFree, Inc.

Sponsor's Authorized Signatory:

Signature/Date: ____

Ephraim Fuchs, MD
Chief Medical Officer
ImmunoFree, Inc.